A Reduced-Form Markov Chain Model of Macro-Financial Regimes

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Abstract

As an alternative to latent state-space modeling, I develop a reduced-form model of macro-financial regimes and apply it to a monthly panel of U.S. macroeconomic and financial time series, spanning 1986 to 2026. I extract orthogonal signals from the series using principal component analysis (PCA), partition the feature space into a discrete state space using k-means clustering, and estimate transition dynamics using empirical transition frequencies. The resulting system has three states, calm, moderate stress, and acute crisis. I report standard Markov-chain quantities, including one and multi-step transition matrices, the stationary distribution, recurrence times, and hitting times, and perform robustness checks to validate the assumption of time homogeneity and the first-order Markov approximation. I find that estimated dynamics accord with economic theory and are consistent with the historical record of financial conditions.

1 Introduction

Economies and financial markets can exhibit markedly different behavior. A long peacetime economic expansion can be punctuated by a financial crisis and sharp recession. A roaring bull market can suddenly crash and take years to recover. Post-crisis malaise can similarly persist. Economists model the evolution of macroeconomic and financial systems in a number of ways. One common approach is to represent a small number of regimes, such as periods of expansion, stress, and acute crisis, as latent states governed by a Markov process, and then infer the state sequence from data. This approach requires specifying a full probabilistic model that links observed variables to latent states, and thus is both complicated and contingent on modeling assumptions.

As an alternative, I present a reduced-form approach in which I construct the state space directly from time series of macro-financial variables and separately estimate the transition dynamics of the resulting discrete process. The time series data characterizes financial conditions. To isolate low-dimensional, orthogonal signals of changing conditions, I use principal component analysis and apply k-means clustering to the resulting score vectors to partition the transformed feature space into a small number of regions. Each region constitutes a regime, and thus the PCA to k-means pipeline defines a deterministic mapping from data to a discrete process which I then model as a time-homogeneous Markov chain.

My objective is primarily methodological, to show that simple tools from stochastic processes and multivariate statistics can obviate joint estimation and produce a reasonable model of macro-financial dynamics. I apply the method to a monthly panel of U.S. macroeconomic and financial variables, spanning January 1986 to March 2026. The result is a three state system, with calm, moderate stress, and crisis regimes. I compute the estimated one and multi-step transition matrices using empirical frequencies, and find that regimes exhibit strong persistence. The crisis regime is rare and short-lived, and transitions to the moderate stress regime. This agrees with economic theory, which suggests that crises follow prolonged deterioration in macro-financial conditions and normalize gradually, rather than arrive suddenly as a result of exogenous shocks. I perform robustness checks for the time-homogeneity assumption and first-order Markov approximation, and conclude they are reasonable.

The paper proceeds as follows. In section 2, I briefly review the academic literature on regime-switching models and applications of dimensionality reduction and clustering methods in macro-finance. In section 3, I formally introduce the Markov chain framework and procedures to estimate transition dynamics. In section 4, I describe the dataset and how I construct features. Section 5 covers the procedure to construct the state space, with a mild treatment of PCA and k-means clustering. In sections 6 and 7, I present the empirical results and validation, and I conclude in section 8 with analysis of my results and a discussion of my method's advantages and limitations.

2 Literature Review

Macroeconomic and financial regime models date at least back to [Hamilton, 1989], which introduced Markov-switching models for macroeconomic time series. Subsequent work, including [Kim and Nelson, 1999], [Ang and Bekaert, 2002], [Chauvet, 1998], [Filardo, 1994], and [Perez-Quiros and Timmermann, 2001], extended these models to multivariate and financial time series. These papers typically specify a parametric observation equation and treat regimes as unobserved states to be inferred using maximum likelihood estimation. I break from this approach by constructing the state space directly from observed data instead of using a fully specified probabilistic model.

Regime-switching models often leverage original time series, and place an emphasis on variable selection. Another set of papers focuses on factor models and the use of principal components to transform and extract the most salient information from macroeconomic and financial time series. For example, [Stock and Watson, 2002] shows that a small number of latent factors can capture the bulk of variation in large macroeconomic datasets, and [Kritzman et al., 2011] shows that principal components can provide interpretable signals of financial conditions. I extend their approach and use PCA as a dimensionality reduction tool to extract orthogonal sources of macro-financial variation. Instead of using the transformed features directly for forecasting or structural interpretation, I use them as inputs to a state construction procedure.

A related strand of the literature takes a different approach to identifying latent states. These papers use clustering and nonparametric methods instead of statistical or structural economic models to construct the state space. [Billio et al., 2010] uses a host of statistical methods to characterize systemic risk and interconnections in financial markets, while more recent work such as [Akioyamen et al., 2021] applies clustering techniques to identify latent market states from high-dimensional data, and in doing so demonstrates that separating the problem of state identification from modeling transition dynamics can provide an interpretable reduced-form characterization. I leverage the idea that regimes can be viewed as regions of the feature space, and then model the temporal evolution of these regimes using a finite-state Markov chain.

Ultimately, this paper does not break new methodological ground. The central contribution is to combine existing ideas from different strands of the academic literature into a cohesive pipeline and demonstrate an alternative to latent state modeling for macro-financial regimes, with applications to real data.

3 Markov Chain Model

A finite-state discrete-time Markov chain provides an interpretable representation of an economy and financial system that transitions between regimes period-to-period [Hamilton, 1989]. In this section, I provide background on the DTMC, following [Durrett, 2016][1].

Formally, let $\{X_t\}_{t\geq 0}$ denote a discrete-time stochastic process taking values in a finite-state space $\mathcal{S} = \{0, 1, \dots, s-1\}$, where s is the number of states and each state corresponds to a macro-financial regime (e.g. calm, transitional, crisis) that I subsequently characterize qualitatively using observed features, and time t is monthly. I model $\{X_t\}$ as a time-homogeneous Markov chain[2] with

$$p(i,j) = p_{ij} = \mathbb{P}(X_{t+1} = j \mid X_t = i), \qquad i, j \in \mathcal{S}.$$

and transition matrix

$$P = (p(i,j))_{i,j\in\mathcal{S}},$$

satisfying $p(i,j) \geq 0$ and $\sum_{j\in\mathcal{S}} p(i,j) = 1$ for all $i \in \mathcal{S}$. The Markov property states that

$$\mathbb{P}(X_{t+1} = j \mid X_t = i, X_{t-1} = i_{t-1}, \dots, X_0 = i_0) = p(i,j),$$

meaning conditional on the present regime, the evolution of the process is independent of the past. Any model of the process makes simplifying assumptions, and in this case the first-order Markov chain is an approximation of true dynamics. The extent to which the Markov property holds is ultimately an empirical question, thus I perform robustness checks in Section 7. To estimate the transition matrix, I use the empirical transition frequency [Leskelä and Dreveton, 2026]. Let $\{x_t\}_{t=1}^{n}$ denote the observed sequence of fitted regime labels, where n is the number of monthly observations. The one-step transition counts are

$$N_{ij} = \sum_{t=1}^{n-1} \mathbf{1}\{x_t = i,\, x_{t+1} = j\}, \qquad N_i = \sum_{j\in\mathcal{S}} N_{ij}.$$

[1] The choice of a finite state space and discrete time reflects two considerations. First, a countably infinite state space diminishes the economic interpretation of any individual state. Markets may broadly appear calm, stressed, or in a crisis, but subdividing aggregated states makes it harder to differentiate one from another. For example, a recovery period can be arbitrarily divided based on recovery speed, but at a certain point, a marginally faster recovery is economically indistinguishable from adjacent states. The state space is thus finite, and Section 5 details the state space construction procedure. Second, because I model regime evolution, not timing, I use discrete time. Macroeconomic regimes are not defined at infinitesimally small time scales, and various economic and financial indicators change value at fixed points. For example, the business cycle evolves over months and years, and is defined by monthly or quarterly publications of GDP growth, unemployment, and changes in the price level. A continuous time chain implies exponentially distributed holding times and regime shifts at random instants within periods [Ross, 2014], which is better suited for event-driven processes rather than the evolution of macro-financial conditions.

[2] In latent state models, $\{X_t\}$ is an unobserved random variable. Here, it is a deterministic function of observed features obtained via clustering. I construct $\{X_t\}$ based on the data, obviating latent state filtering.

The transition probability estimator is the empirical frequency of transitions from states i to j,

$$\hat{p}(i,j) = \frac{N_{ij}}{N_i}, \qquad N_i > 0.$$

By construction, $\hat{P}$ is row-stochastic and, under standard regularity conditions, a consistent estimator of P. The estimated transition matrix $\hat{P}$ captures both the persistence of regimes and the probability of transitions between them. High diagonal entries indicate strong persistence, while off-diagonal entries reflect the frequency and direction of regime shifts. Transition dynamics reflect economic evolution and the geometry of the clustering procedure to construct the state space.

Standard long-run results for a finite-state Markov chain apply when the chain is irreducible and aperiodic[3]. A probability vector $\pi = (\pi_0, \ldots, \pi_{s-1})$ is stationary if

$$\pi = \pi P, \quad \sum_{i \in \mathcal{S}} \pi_i = 1.$$

If the chain is irreducible and aperiodic on $\mathcal{S}$, there exists a unique stationary distribution that governs its long-run behavior. Specifically, as time goes to infinity, the probability of the system being in regime j converges to π_j, regardless of where it started, thus

$$\lim_{h \to \infty} P^h(i,j) = \pi_j \qquad \text{for all } i, j \in \mathcal{S}.$$

I compute the stationary distribution $\hat{\pi}$ associated with $\hat{P}$ by solving $\hat{\pi} = \hat{\pi}\hat{P}$ and compare $\hat{\pi}$ with the empirical state frequencies

$$\tilde{\pi}_i = \frac{1}{n} \sum_{t=1}^{n} \mathbf{1}\{x_t = i\},$$

as a robustness check. I use the stationary distribution and $\hat{P}$ to compute several measures.

First, the mean recurrence time of a state represents the expected time between successive visits to that state, counting a one-period self-transition as a return. For $i \in \mathcal{S}$, define the first return time to state i by

$$T_i^+ = \inf\{t \geq 1 : X_t = i\}.$$

The mean recurrence time is

$$r_i = \mathbb{E}_i[T_i^+],$$

where $\mathbb{E}_i[\cdot]$ denotes expectation conditional on $X_0 = i$. For a finite irreducible chain,

$$r_i = \frac{1}{\pi_i}.$$

[3]Whether these properties hold is ultimately an empirical question, but I argue they are at least plausible. First, the state space represents recurrent phases of a single macro-financial system, thus over sufficiently long horizons, states should communicate with each other. Second, there is no reason to expect deterministic cycles in regime transitions, and there is good reason to expect positive self-transition probabilities based on regime persistence.

I estimate mean recurrence times using

$$\hat{r}_i = \frac{1}{\hat{\pi}_i}.$$

Crisis regimes should be less frequent than stable regimes, thus I expect larger recurrence times for the former compared to the latter.

Second, the expected duration of time the chain spends in a given regime. While in state i, the chain remains there with probability $p(i,i)$ and leaves with probability $1 - p(i,i)$. For state $i \in \mathcal{S}$, define the first exit time from state i by

$$D_i = \inf\{t \geq 1 : X_t \neq i\}.$$

Conditional on $X_0 = i$, D_i follows a geometric distribution with success probability $1 - p(i,i)$, thus

$$\mathbb{E}_i[D_i] = \frac{1}{1 - p(i,i)},$$

which I interpret as the persistence of each macro-financial regime.

Third, the expected hitting time. For $i, j \in \mathcal{S}$, the first hitting time of state j starting from i is

$$\tau_j = \inf\{t \geq 0 : X_t = j\}.$$

I solve for the expected hitting times $m_{ij} = \mathbb{E}_i[\tau_j]$ using the system

$$m_{ij} = \begin{cases} 0, & i = j, \\ 1 + \sum_{k \in \mathcal{S}} p(i,k) m_{kj}, & i \neq j. \end{cases}$$

and the estimated transition matrix $\hat{P}$[4].

Fourth, the h-step transition probabilities,

$$p^{(h)}(i,j) = \mathbb{P}(X_{t+h} = j \mid X_t = i) = P^h(i,j),$$

with empirical analogues

$$\hat{p}^{(h)}(i,j) = \hat{P}^h(i,j).$$

For an irreducible and aperiodic finite-state chain, $P^h(i,j) \to \pi_j$ as $h \to \infty$, and these probabilities provide a forward-looking view of regime dynamics. Discrete-time, finite-state Markov chains have a deep theoretical grounding, and there are more quantities to report. I focus on the most useful objects from the perspective of validating and using the model, which provide a thorough description of regime dynamics, persistence, transition behavior, and the relative prevalence of the different macro-financial states.

[4] Expected hitting times characterize the accessibility and rarity of regimes under the estimated dynamics. For example, a crisis regime should be difficult to reach from calm conditions, but once reached should not be absorbing because markets eventually do stabilize [Danielsson et al., 2016].

4 Data and Feature Construction

Macro-financial regimes are defined by the joint variation in a range of variables [Ha et al., 2020]. Accordingly, I construct a monthly panel of macroeconomic and financial variables to capture both cyclical conditions and financial stress, including shifts in growth, inflation dynamics, the shape of the yield curve, equity market levels and volatility, and credit risk. Instead of imposing a regime structure by specifying deterministic criteria for select variables, I purposely favor a broad set of variables to provide a rich feature space[5].

The sample begins in January 1985[6] and extends through the latest available observation (March 2026 as of writing). I convert all series to a common monthly frequency. For daily series such as VIX, which measures equity market option-implied volatility [Cboe Global Markets, 2026], and the St. Louis Fed Financial Stress Index, a derived series which measures the degree of financial stress in markets [Federal Reserve Bank of St. Louis, 2026b], I take monthly averages. For all other series, I use end-of-month values. I obtain all data from FRED [Federal Reserve Bank of St. Louis, 2026a] and Yahoo Finance [Yahoo Finance, 2026] using standard APIs. The raw series, particularly the interest rate level variables and macroeconomic aggregates, exhibit substantial persistence, thus I apply standard transformations to mitigate unit-root behavior. See Appendix A for details. After aligning all variables and removing missing observations, the final balanced panel used for PCA contains 481 monthly observations of 12 variables (S&P 500, CPI, industrial production, unemployment rate, term spreads, credit spreads, spread changes) spanning January 1986 to March 2026 (inclusive). Table 1 summarizes.

Table 1: PCA Feature Variables (Jan. 1986 – Mar. 2026, Monthly, n=481)

Variable	Tickers	Source	Interpretation	Transformation
r_t^{SP500}	GSPC	YF	Equity returns	Monthly return
π_t (CPI)	CPIAUCSL	FRED	Inflation	YoY change
g_t^{ip}	INDPRO	FRED	Output growth	YoY change
Δu_t	UNRATE	FRED	Unemployment change	First difference
$s_t^{10y,3m}$	GS10, TB3MS	FRED	Term spread	Level
$\Delta s_t^{10y,3m}$		(derived)	Change in term spread	First difference
$s_t^{10y,2y}$	GS10, GS2	FRED	Term spread	Level
$\Delta s_t^{10y,2y}$		(derived)	Change in term spread	First difference
$c_t^{baa,aaa}$	BAA, AAA	FRED	Credit spread	Level
$\Delta c_t^{baa,aaa}$		(derived)	Change in credit spread	First difference
$c_t^{baa,10y}$	BAA, GS10	FRED	Credit spread	Level
$\Delta c_t^{baa,10y}$		(derived)	Change in credit spread	First difference

[5]Allowing the data to in effect reveal the state space is a convenient alternative to principled variable selection, which requires a formal economic model and is beyond the scope of this paper.

[6]The raw sample begins in January 1985. Due to lagged transformations (e.g. YoY percent, differences), the transformed sample begins in January 1986.

I purposefully exclude VIX, the St. Louis Fed Financial Stress Index, and the NBER Recession Indicator (defined as a significant decline in economic activity spread across the economy and lasting more than a few months [National Bureau of Economic Research, 2026]) from the PCA feature set. First, these variables comprise the validation set, which I use to assess and interpret constructed regimes. To be of any use, they cannot be inputs. Second, these variables are purpose-built to define macro-financial regimes[7], thus their inclusion is antithetical to the goal of *discovering* regimes.

Table 2: Summary Statistics for PCA Features (Jan. 1986 – Mar. 2026, Monthly, n=481)

Variable	Mean	Std.	Min.	Median	Max.	Skew
r_t^{SP500}	0.81	4.38	-21.76	1.22	13.18	-0.71
π_t (CPI)	2.79	1.58	-1.96	2.67	8.98	0.85
g_t^{ip}	1.63	4.04	-17.32	2.25	16.55	-1.39
Δu_t	-0.01	0.53	-2.20	0.00	10.40	15.92
$s_t^{10y,3m}$	1.56	1.21	-1.57	1.58	3.76	-0.19
$s_t^{10y,2y}$	0.97	0.89	-0.93	0.84	2.83	0.26
$c_t^{baa,aaa}$	0.96	0.36	0.51	0.90	3.38	3.02
$c_t^{baa,10y}$	2.28	0.69	1.29	2.16	6.01	1.80
$\Delta s_t^{10y,3m}$	-0.00	0.23	-0.98	-0.01	0.80	0.16
$\Delta s_t^{10y,2y}$	-0.00	0.14	-0.72	-0.01	0.59	0.04
$\Delta c_t^{baa,aaa}$	-0.00	0.10	-0.63	-0.01	0.94	1.59
$\Delta c_t^{baa,10y}$	-0.00	0.17	-0.99	-0.01	1.45	1.72

Table 2 reports summary statistics for the feature variables, where equity returns, inflation, and output growth are percentages, while the monthly change in the unemployment rate, Δu_t, is in percentage points. Monthly S&P 500 returns have a mean of 0.8% and standard deviation of 4.4% (10% and 15% annualized, respectively), and exhibit negative skewness. Inflation and industrial production growth have means of 2.8% and 1.6%, respectively, consistent with the higher output and stable price growth of long-run economic expansions. Industrial production is more volatile because it is pro-cyclical, with the largest decreases (e.g. -17%) occurring during recessions. The change in the unemployment rate is on average 0. This indicates that there is no systematic upward or downward drift in the unemployment rate in levels over the sample. The positive skewness and the observed maximum of 10.4 percentage points reflect unemployment spikes during recessions

[7] This is obvious for the recession indicator, which is an official, binary, business cycle label (1 for recessions and 0 for expansionary macroeconomic periods) and thus conceptually distinct from the other series. Similarly, the STLFSI is itself the first principal component of 18 variables measuring financial stress [Kliesen and McCracken, 2020], and its inclusion is in effect double counting. Including STLFSI overweights on financial stress rather than broader macro-financial stress. VIX is forward-looking option-implied equity volatility. It is commonly interpreted as a fear gauge, reflecting investor sentiment and risk aversion. VIX is problematic because it is high-frequency and very reactive, especially during stress periods (e.g. GFC, Covid) where its spikes can dominate slower moving macroeconomic variables and thus the component structure. For example, some papers explicitly bin on VIX quantiles to classify regimes, and I define states as broad macro-financial regimes, not merely equity market conditions.

(e.g. Covid). In levels, the term and credit spreads are positive on average. Spread volatility is low. Historically, the yield curve is upward sloping and bonds that are less liquid and bear default risk have higher yields than comparable US Treasuries [Gilchrist and Zakrajšek, 2012]. First-differenced spreads are on average 0 with minimal variance, again indicating no systematic drift in levels over the sample. The maximum monthly changes in spreads are comparable to the levels averages, confirming that first-differenced spreads do capture short-run fluctuations.

Following [Stock and Watson, 2002], I perform dimensionality reduction using principal component analysis. PCA assumes finite second moments and a well-behaved covariance structure, thus I examine the time-series stationarity of the variables using complementary Augmented Dickey–Fuller (ADF) and KPSS tests[8]. Table 3 reports the results.

Table 3: Stationarity Tests for PCA Features

Variable	ADF Stat	ADF p	ADF Lags	KPSS Stat	KPSS p	Classification
r_t^{SP500}	-16.3784	0.0000	1	0.0885	0.1000	Stationary
π_t (CPI)	-3.5001	0.0080	15	0.3099	0.1000	Stationary
g_t^{ip}	-3.7375	0.0036	14	0.5220	0.0367	Inconclusive
Δu_t	-12.9081	0.0000	3	0.0299	0.1000	Stationary
$s_t^{10y,3m}$	-3.9334	0.0018	8	0.6104	0.0217	Inconclusive
$s_t^{10y,2y}$	-3.6587	0.0047	8	0.2647	0.1000	Stationary
$c_t^{baa,aaa}$	-4.9211	0.0000	5	0.1761	0.1000	Stationary
$c_t^{baa,10y}$	-3.7049	0.0040	2	0.4441	0.0581	Stationary
$\Delta s_t^{10y,3m}$	-10.6618	0.0000	2	0.0301	0.1000	Stationary
$\Delta s_t^{10y,2y}$	-5.2117	0.0000	7	0.0451	0.1000	Stationary
$\Delta c_t^{baa,aaa}$	-8.8924	0.0000	7	0.0220	0.1000	Stationary
$\Delta c_t^{baa,10y}$	-13.8953	0.0000	1	0.0265	0.1000	Stationary

Aside from industrial production growth and the 10Y–3M term spread, all variables are stationary per both tests. For both, the ADF test rejects a unit root, but the KPSS test rejects stationarity. Either null hypothesis is plausible. Industrial production growth is naturally bounded by overall economic growth, and thus should not exhibit a linear time trend, but it is especially volatile around recessions, and thus may have a time-varying second moment. Similarly, the term spread is bounded by the no arbitrage pricing relationship among bonds of different maturities, but empirically it can switch signs (e.g. yield curve inversion) during inflation cycles and recessions. Despite inconclusive test results, I retain both because they convey critical macroeconomic information. Overall, the time-series properties of the transformed features indicate they are suitable for PCA. See Appendix B for complete diagnostics, including tests for conditional heteroskedasticity and structural breaks.

[8]The ADF null is that the series is I(1), thus the test asks whether a unit root can be rejected. The KPSS null is that the series is stationary around a constant, meaning KPSS asks whether stationarity can be rejected. I test at $\alpha = 5\%$ and specify a constant in the ADF regression. For both, I select lag order using AIC and implement the tests with `statsmodels` [Statsmodels Developers, 2025].

5 State Space Construction

Each state in the Markov chain corresponds to a macro-financial regime. The crux of the model is thus the procedure to construct the state space, because the estimated transition matrix and all associated dynamics are contingent on the state space. I begin with time series of macro-financial variables, Y_t. I apply PCA to this high-dimensional feature set and obtain low-dimensional principal score vectors, Z_t[9]. The eigenvectors are orthogonal, and thus the scores represent unique, mutually uncorrelated (they may be serially correlated) signals of macro-financial conditions. Regimes are discrete labels, thus I use k-means clustering to partition the continuous feature space into groups of similar macro-financial conditions. I interpret the resulting clusters, C_k, as distinct regimes following [Billio et al., 2010], and define the process X_t based on which cluster month t falls in. Concisely, the pipeline is $Y_t \to Z_t \to C_k \to X_t$.

5.1 Principal Component Analysis

I review principal component analysis (PCA) following [Jolliffe, 2002] and report results. Let $Y_t \in \mathbb{R}^p$ denote the vector of transformed features at time t, where $p = 12$. I standardize each feature to have zero mean and unit variance, obtaining $\widetilde{Y}_t$, and compute the sample covariance matrix,

$$\Sigma = \frac{1}{n-1} \sum_{t=1}^{n} \widetilde{Y}_t \widetilde{Y}_t',$$

PCA uses the spectral decomposition of Σ. Since Σ is symmetric PSD, it can be decomposed as

$$\Sigma = V \Lambda V',$$

where $V = (v_1, \dots, v_p)$ is an orthogonal matrix whose columns are eigenvectors, and $\Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_p)$ is a diagonal matrix of eigenvalues satisfying $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p \geq 0$. For orthonormal eigenvectors,

$$v_i' v_j = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

The ℓ-th principal component is the eigenvector $v_\ell \in \mathbb{R}^p$, which defines a linear combination of the features (geometrically, a direction in feature space) and can be interpreted using factor loadings (elements of v_ℓ). Factor loadings provide an economic interpretation of each component, indicating how the underlying macro-financial variables contribute to each source of variation. The ℓ-th principal component score is the projection of the data onto the ℓ-th eigenvector,

$$Z_{t,\ell} = v_\ell' \widetilde{Y}_t.$$

[9]Macro-financial variables are highly correlated (See Appendix C for a correlation matrix). I start with high-dimensional data to capture as many sources of variation as possible, and then rely on orthogonalization from PCA to avoid double counting economic information when constructing clusters. Tying the definition, and thus interpretability and stability, of the regime to the geometry of the feature space requires PCA [Kritzman et al., 2011].

By construction, the principal component scores are uncorrelated,

$$\mathrm{Cov}(Z_{t,i}, Z_{t,j}) = 0 \quad \text{for } i \neq j,$$

and have variances given by the corresponding eigenvalues,

$$\mathrm{Var}(Z_{t,\ell}) = \lambda_\ell.$$

The eigenvalues quantify the amount of variation explained by each component. The proportion of variance explained (PVE) by the ℓ-th component is

$$\mathrm{PVE}_\ell = \frac{\lambda_\ell}{\sum_{j=1}^{p} \lambda_j},$$

and the cumulative PVE summarizes how much of the total variation is captured by the first m components. The purpose of PCA is to obtain a lower-dimensional representation of the feature space that captures the maximal amount of variation in the original data, thus I select $m \ll p$. Table 4 reports the proportion of variance explained by each component.

Table 4: Explained Variance by Principal Component

Component	Explained Variance	Cumulative Variance
PC1	0.2556	0.2556
PC2	0.1671	0.4227
PC3	0.1404	0.5631
PC4	0.1269	0.6901
PC5	0.0860	0.7761
PC6	0.0693	0.8455
PC7	0.0607	0.9061
PC8	0.0329	0.9390
PC9	0.0295	0.9686
PC10	0.0148	0.9834
PC11	0.0124	0.9958
PC12	0.0042	1.0000

The first four principal components explain approximately 69% of the total variance, while the first five account for over 77%. Based on the diminishing marginal contribution to cumulative percent of variance explained, I select $m = 4$ and use PC1, PC2, PC3, and PC4's scores to construct regimes. Table 5 reports the top loadings by magnitude for the first four principal components. See Appendix D for the full loading matrix. Principal components are linear combinations of variables, thus the magnitude of the coefficients indicate which variables the PCs load on and their relative importance to the component. Although the signs are arbitrary, in the sense that multiplying the component by -1 gives the same statistical content, relative signs are meaningful. Opposite signs mark a constrast factor, whereas the same signs mark co-movement.

Table 5: Top Loadings for Principal Components 1 – 4 (Magnitude)

Rank	PC1	PC2	PC3	PC4
1	$c_t^{baa,10y}$ (0.5080)	$\Delta c_t^{baa,aaa}$ (0.6076)	$\Delta s_t^{10y,2y}$ (-0.5527)	$\Delta s_t^{10y,3m}$ (-0.4347)
2	$c_t^{baa,aaa}$ (0.4335)	$\Delta c_t^{baa,10y}$ (0.6025)	$\Delta s_t^{10y,3m}$ (-0.5111)	$\Delta s_t^{10y,2y}$ (-0.3847)
3	$s_t^{10y,2y}$ (0.4058)	Δu_t (0.2327)	$s_t^{10y,3m}$ (0.4148)	g_t^{ip} (-0.3577)
4	g_t^{ip} (-0.3537)	r_t^{SP500} (-0.2200)	$s_t^{10y,2y}$ (0.3474)	$s_t^{10y,3m}$ (-0.3515)
5	$s_t^{10y,3m}$ (0.3401)	π_t (0.1871)	g_t^{ip} (0.3227)	$s_t^{10y,2y}$ (-0.3115)
6	π_t (-0.2948)	$c_t^{baa,aaa}$ (0.1819)	Δu_t (-0.1197)	r_t^{SP500} (0.3018)

I use the factor loadings to interpret PC1-PC4, following the reasoning of [Estrella and Mishkin, 1998] and [Gilchrist and Zakrajšek, 2012].

- **PC1 (Financial Conditions Factor):** PC1 loads positively on credit spreads ($c_t^{baa,10y}$, $c_t^{baa,aaa}$) and term spreads ($s_t^{10y,2y}$, $s_t^{10y,3m}$), and negatively on real activity measures such as g_t^{ip} and π_t. Loading magnitudes appear uniform. PC1 is higher when the credit risk premium is higher and the yield curve is upward sloping, and lower when real economic activity and inflation are strong. PC1 primarily signals financial conditions in levels, specifically risk premia and curve steepness, and the loadings suggest PC1 is high in early recovery or post-stress environments and low in inflationary or tightening monetary policy environments.

- **PC2 (Credit-Labor Interaction Factor):** PC2 loads positively on changes in credit spreads ($\Delta c_t^{baa,aaa}$, $\Delta c_t^{baa,10y}$) and the change in the unemployment rate (Δu_t). The magnitudes for loadings on first-differenced credit spreads are three times the magnitudes for other variables, thus PC2 primarily captures short-run *changes* in credit market conditions. Positive exposure to widening credit spreads, increasing unemployment, and the negative loading on equity returns suggest PC2 reflects the onset or intensifying of stress episodes.

- **PC3 (Financial Adjustment Factor):** PC3 loads negatively on changes in term spreads ($\Delta s_t^{10y,2y}$, $\Delta s_t^{10y,3m}$) but positively on term spreads in levels, albeit with lower magnitudes. The positive exposure to industrial production growth (g_t^{ip}) and negative exposure to changes in the unemployment rate suggests PC3 is sensitive to underlying economic conditions, but primarily reflects rapid repricing in the bond market, consistent with investors changing expectations as financial conditions adjust.

- **PC4 (Decoupling Factor):** PC4 loads negatively on industrial production, term spreads in levels, and first-differenced spreads, all with comparable magnitudes. It is the only factor with large, positive exposure to equity returns (r_t^{SP500}), thus it captures periods where equities perform well despite a flatter yield curve in levels, curve flattening, and weak industrial production, meaning market performance is decoupled from real activity.

5.2 Clustering

I use clustering to partition the continuous feature space into discrete regions. For cluster region $C_k \subset \mathbb{R}^m$ and month t's scores $(Z_{t1}, Z_{t2}, Z_{t3}, Z_{t4})$, I define the discrete process X_t as

$$X_t = k \quad \text{if } (Z_{t1}, Z_{t2}, Z_{t3}, Z_{t4}) \in C_k, \qquad k = 0, \dots, K-1.$$

Following [Akioyamen et al., 2021], I interpret each cluster as a state and the time series $\{X_t\}$ as the evolution of the system across the states, hence I use the terms state and regime interchangeably.

Although clustering algorithms traditionally operate on cross-sectional data, they only require that observations be represented as vectors in a feature space, and thus can easily be applied to time-series data[10]. The choice of clustering algorithm matters only if it results in materially different regimes, thus I favor k-means because it is simple and interpretable[11]. Let $A_k = \{t : \vec{Z}_t \in C_k\}$ denote the set of time indices assigned to cluster k. Formally, clustering solves

$$\min_{\{C_k\}_{k=0}^{K-1}} \sum_{k=0}^{K-1} \sum_{t \in A_k} \|Z_t - \mu_k\|^2, \quad \mu_k = \frac{1}{|A_k|} \sum_{t \in A_k} Z_t$$

where μ_k is the centroid of cluster C_k. k-means is an iterative algorithm. It first assigns observations to the nearest centroid, measured in Euclidean distance, and then sets the updated centroid to be the mean of points assigned to the cluster. It then repeats this process until the objective function converges to a local minimum [Ikotun et al., 2023]. With sufficient time, convergence is guaranteed.

The primary hyperparameter is K, the number of clusters. The choice of K is fundamentally economic, not statistical. If K is too large, each individual cluster is less differentiated from the others. In the extreme case of one cluster per observation, the objective function reaches its lower bound of 0 but each month trivially becomes a distinct regime. If K is too small, the risk is that economically distinct regimes get merged. I begin with a baseline of $K = 3$, corresponding to calm/expansion, intermediate/moderate stress, and crisis periods, and test $K = 2$ and $K = 4$ for robustness using clustering metrics, the validation variables (VIX, STLFSI, NBER), and resulting regime properties. See Appendix F for details on the procedure, complete results, and analysis.

I find that at $K = 4$, clustering metrics improve modestly, but this may simply reflect k-means mechanically reducing intra-cluster variance as described above. Moreover, the crisis regime gets split into two smaller clusters. There are so few observations for each that estimating transition dynamics is pointless because of the sampling variability. At $K = 2$, the sole stress period collapses

[10] Mechanically, the algorithm will output clusters regardless of whether the inputs are time series or cross-sectional. Substantively, it is true that k-means ignores temporal dependence by treating $Z_{t,1} \dots Z_{t,4}$ as an unordered collection of points in $\mathbb{R}^4$. For example, Sept. 2008 (GFC) and March 2020 (Covid) may be geometrically close in PCA space, despite being 12 years apart. This is not an issue, because my goal is to group similar environments based on macro-financial conditions, regardless of when they occur, and separately estimate transition dynamics. I model regime persistence and transition behavior ex-post because the Markov property states that $\mathbb{P}(X_{t+1} = j \mid X_t, X_{t-1}, \dots) = \mathbb{P}(X_{t+1} = j \mid X_t)$, and the only temporal dependence in the sequence $\{X_t\}$ stems from the current state.

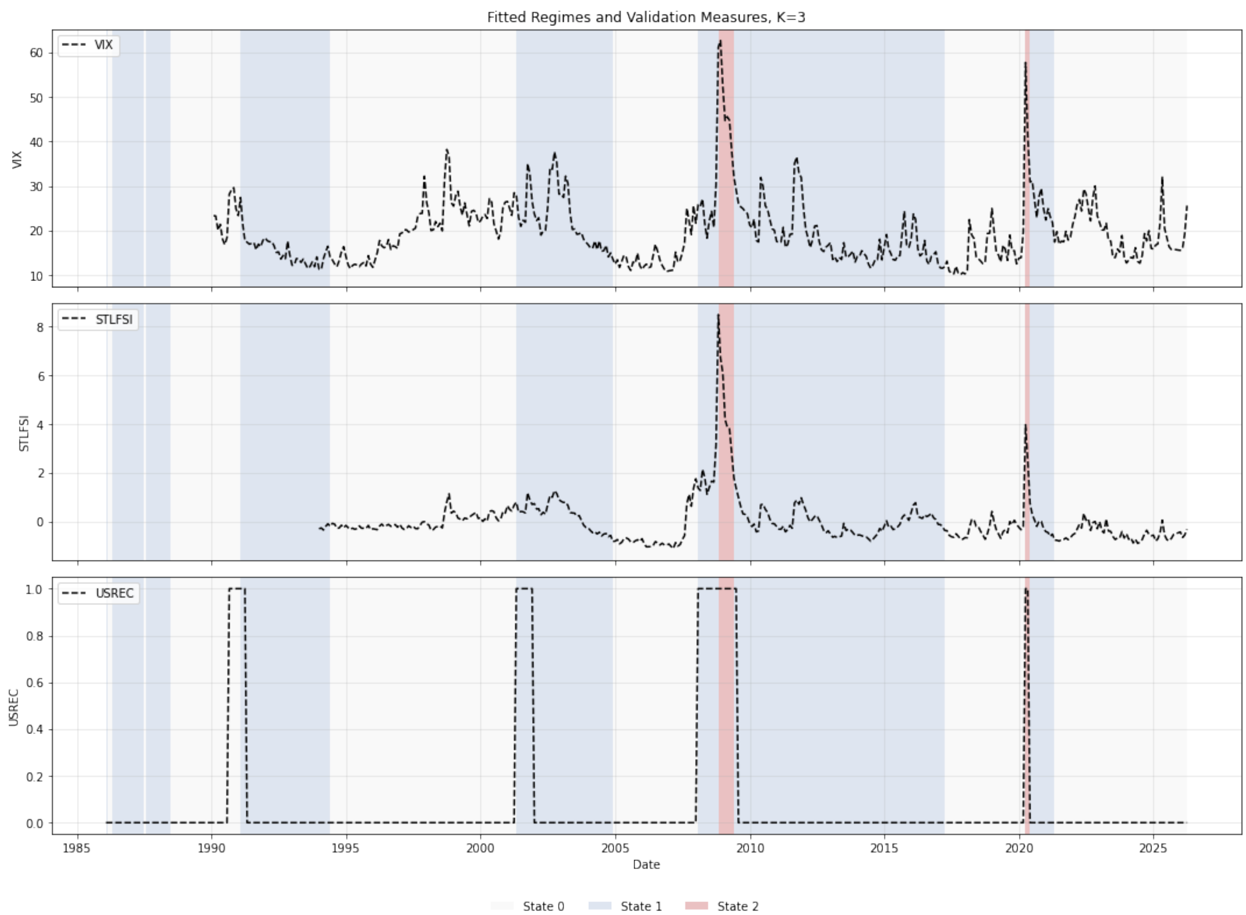
[11] It does assume spherical clusters and equal variance, both of which are unlikely for macro-financial regimes. This is not an issue because if there is a distortionary effect, it will occur at the regime boundary and thus have a minimal impact on transition dynamics.

the distinction between moderate stress and acute crisis. Events like the GFC and Covid appear in the same regime as the prolonged recoveries from those events. $K = 3$ has comparable statistical fit and is the most compelling economically, with two dominant regimes, a rare crisis regime, and its periods neatly align with VIX, STLFSI, and the NBER Recession Indicator. See Appendix E for cluster fits. Table 6 shows regime shares and durations.

Table 6: Regime Shares and Durations

State	Interpretation	Count	Share	Avg. Duration (months)
0	Calm / Expansion	249	51.77%	35.57
1	Moderate Stress	223	46.36%	27.88
2	Crisis	9	1.87%	4.50

State 0 comprises 52% of the sample, State 1 comprises 46%, and State 2 comprises 2%. Figure 1 plots regimes (shaded) against the VIX, STLFSI, and NBER time series.



Based on the regime shares, durations, and context from the validation measures, the states are consistent with stylized macro-financial facts. State 0 corresponds to unshaded periods, and is calm, with low volatility and near-zero recession probability, and empirically coincides with macroeconomic expansions. It lasts on average 3 years. State 1 is a stressed regime, with moderate volatility, elevated credit risk premiums, and an increased probability of a recession. It lasts on average just over 2 years and appears during extended periods of elevated macro-financial risk, including the early 1990s recession, the dot-com recession, the pre-GFC period, and the long period of subdued but nonzero stress following the financial crisis. State 2 is a crisis regime, marked by extreme volatility (VIX over 50), high financial stress, and recessions. It is much rarer and aligns with acute episodes, most clearly the 2008–2009 financial crisis and the Covid shock in 2020 [National Bureau of Economic Research, 2026].

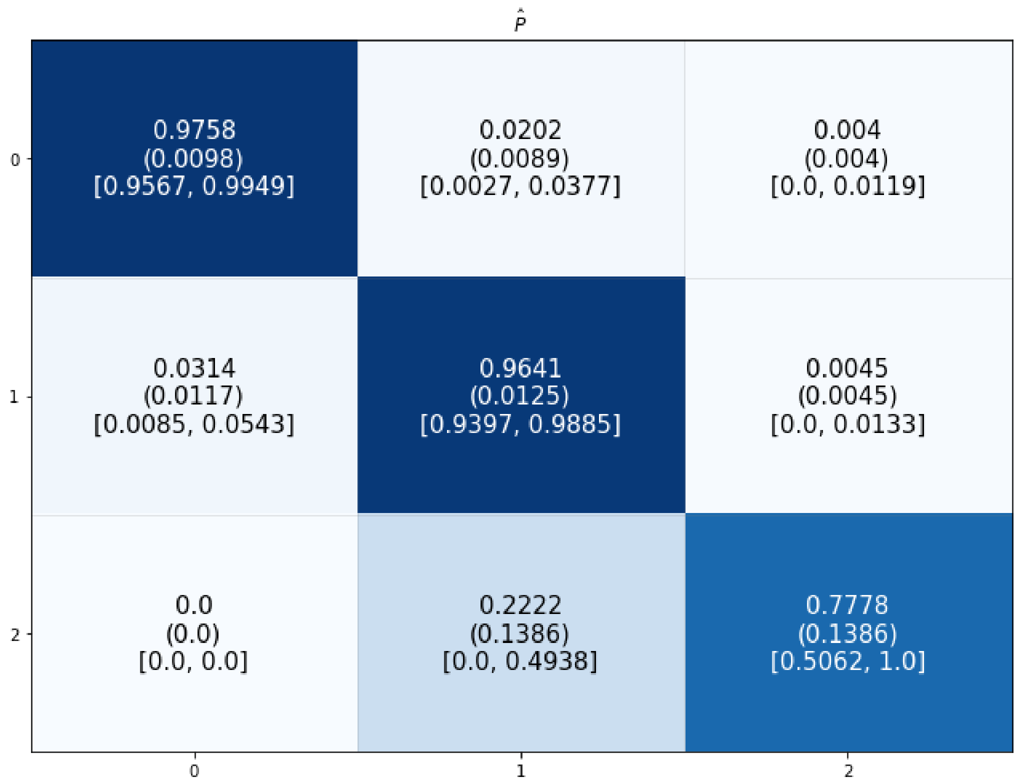
State space construction using PCA and k-means is the core of the Markov chain. Having established the mapping from observed data to discrete regimes, I estimate the transition dynamics and complete the remaining structure of the model[12][13].

[12]The resulting regimes depend on modeling choices, such as the number of clusters, PCA scaling and dimension, and choice of sample period. I conduct robustness checks to address sensitivity of the fitted regimes to the choices, but identification is primarily economic. I begin with the prior of three intuitive states and find that neither different modeling choices made nor observed evidence dislodges that prior.

[13]Before using the score vectors in clustering, I repeat the time-series tests. All score vectors exhibit serial correlation and conditional heteroskedasticity. This is to be expected given both are present in the underlying features. PC3 and PC4 exhibit some evidence of a structural break. PC1 and PC3 are clearly stationary, but the ADF and KPSS split results for PC2, and PC4 is nonstationary.

6 Transition Dynamics

As stated in Section 3, the state space and transition matrix completely specify a DTMC, and transition dynamics follow from the regime sequence $\{X_t\}$. I estimate the transition probability matrix using the empirical frequency estimator, and show the results in Figure 2. Each cell contains the estimated transition probability $\hat{p}(i, j)$, with the approximate standard error in parentheses and the 95% confidence interval in brackets[14]. Darker cells indicate larger transition probabilities.



Three points stand out. First, the regime structure is highly persistent. For each state, diagonal entries are large, with $\hat{p}_{00} = 0.9758$, $\hat{p}_{11} = 0.9641$, and $\hat{p}_{22} = 0.7778$. The calm and moderate-stress regimes are especially stable, while the crisis state is shorter lived but still persistent once entered.

Second, off-diagonal entries show a pattern for regime change. From the calm state, the probability of moving to moderate stress is $\hat{p}_{01} = 0.0202$, while the probability of moving directly to crisis is only $\hat{p}_{02} = 0.0040$. If the system leaves a calm period, it is much more likely to transition to moderate stress than crisis. This result is consistent with the view that crises often emerge after a deterioration in financial conditions, rather than appearing immediately from otherwise calm environments [Adrian et al., 2019]. From moderate stress, the probability of returning to calm is

[14] Standard errors and confidence intervals follow from a multinomial approximation. Per [Anderson and Goodman, 1957], the transition counts conditional on state i are multinomial with sample size N_i, and thus the variance of $\hat{p}_{ij}$ is approximately $\hat{p}_{ij}(1 - \hat{p}_{ij})/N_i$. From the variance I compute standard errors and construct Wald intervals $\hat{p}_{ij} \pm 1.96 \cdot \mathrm{SE}(\hat{p}_{ij})$, truncating them to $[0, 1]$ so that the interval bounds are valid probabilities. Because the distribution is approximate, the quality naturally degrades with smaller samples.

$\hat{p}_{10} = 0.0314$, while the probability of moving to crisis is only $\hat{p}_{12} = 0.0045$. Moderate stress periods are roughly seven times more likely to normalize than to deteriorate into a crisis, thus it should not be interpreted as a deterministic *pre*-crisis state, but rather as a persistent intermediate regime in which financial stress conditions are elevated but usually resolve without crisis. Indeed, Figure 1 shows it often *follows* recessions.

Third, there are no observed direct transitions from crisis to calm in one month, whereas the estimated probability of moving from crisis to moderate stress is $\hat{p}_{21} = 0.2222$. This indicates that when crises end, markets do not immediately revert to calm conditions, but rather normalize gradually. Again, the model coheres with economic intuition. Acute stress periods tend to leave behind impaired credit markets, elevated volatility, and weak macroeconomic conditions (higher unemployment, lower output) even after the immediate crisis has passed[15] [Ha et al., 2020].

Figure 3 shows the system. Although the obvious implication is that crises are rare events that follow from an accumulation of financial stress, and thus shocks like Covid or the GFC tip a fragile system into acute crisis, the model does not exclude the case of purely exogenous shocks disrupting a calm regime. It simply posits that such a case is rare relative to the case of gradual deterioration, making the system vulnerable. The chain is finite, irreducible, and aperiodic. All probabilities, except for $\hat{p}_{20}$, are strictly positive, thus all states have positive transition probability. All states communicate with each other either directly or indirectly. For example, although $\hat{p}_{20}$ is 0, the chain can move $2 \to 1$ and $1 \to 0$. Each diagonal entry of the matrix is strictly positive, thus the system can self-loop at each state, and the chain is aperiodic. Given irreducibility and aperiodicity, there exists a unique stationary distribution to which the long-run multi-step transition probabilities converge to. Table 7 reports empirical and model-implied regime characteristics, including the stationary distribution, expected duration, and mean recurrence time.

Table 7: Model-Implied and Empirical Regime Dynamics

	Model-Implied				Empirical	
State	$\hat{p}_{ii}$	$\hat{\pi}_i$	$\widehat{\mathbb{E}}_i[D_i]$	$\widehat{\mathbb{E}}_i[T_i^+]$	Share	Avg. Duration
0	0.9758	0.5542	41.33	1.80	51.77%	35.57
1	0.9641	0.4271	27.88	2.34	46.36%	27.88
2	0.7778	0.0187	4.50	53.55	1.87%	4.50

January 1986 – March 2026 (monthly, $n = 481$). State 0 = Calm (249), State 1 = Moderate Stress (223), State 2 = Crisis (9). Model-implied quantities follow from estimated transition matrix $\hat{P} = (\hat{p}(i,j))_{i,j\in\mathcal{S}}$. Diagonal $\hat{p}_{ii}$ is probability of remaining in state i. Stationary distribution $\hat{\pi}_i$ satisfies $\hat{\pi} = \hat{\pi}\hat{P}$. Expected duration is estimated as $\widehat{\mathbb{E}}[D_i] = 1/(1-\hat{p}_{ii})$, where D_i is the first exit time from state i. Mean recurrence time is estimated as $\widehat{\mathbb{E}}_i[T_i^+] = 1/\hat{\pi}_i$, where T_i^+ is the first return time to state i starting from i. Empirical quantities computed from observed regime sequence. Shares denote sample frequency. Average duration is mean observed run length. All time units in months. States zero-indexed for code (Python) consistency.

[15] Calm and moderate regime estimates are precise, reflecting the large number of observations. The crisis state only has 9 observations and thus large SEs and a wide CI, so its estimates should be interpreted cautiously.

The comparison is revealing. First, the stationary distribution closely matches the empirical state frequencies, indicating that the model captures the dominant time allocation across regimes. The crisis state even has an identical stationary and empirical share[16]. Second, the calm and moderate states have high stationary probabilities and thus short recurrence times of approximately 1.8 and 2.3 months, respectively. The system spends years in each regime, with model-implied expected durations of 41 and 28 months, respectively. The crisis state is comparatively rare, with a low stationary probability and a mean recurrence time of 54 months.

I next compute expected hitting times. For states i and j, the expected hitting time m_{ij} is the expected number of months required to first reach state j starting from state i.

Table 8: Expected Hitting Times Across Regimes

Starting State	To State 0	To State 1	To State 2
0	0.00	42.08	238.38
1	32.50	0.00	236.46
2	37.00	4.50	0.00

From a calm state, the expected time to reach a moderate stress period is 42 months, and to a crisis period is 238 months. From a period of moderate stress, the expected time to revert to calm is 32.5 months and to reach a crisis is 236 months. The long expected hitting times to the crisis regime reflect low transition probabilities on account of how rare crises are in the sample, and should not be interpreted literally as crises occur once every 20 years. From the crisis state, the expected hitting time to moderate stress is 4.5 months, and to calm is 37 months. The model thus implies crises moderate within a few months, but it takes several years for macro-financial conditions to completely normalize.

Repeatedly multiplying the one-step transition matrix by itself gives multi-step transition probabilities. I compare the h-step transition matrix $\widehat{P}^h$ to empirical multi-step transition probabilities, again computed using the empirical frequency estimator. Appendix G reports complete results at $h = 2, 3, 6, 12$. The 12-month transition dynamics show the same patterns as the 1-month dynamics. At one year, the probability of remaining in a calm regime is approximately 0.78, while the probability of moving from calm to intermediate stress is approximately 0.20. The probability of going directly from calm to crisis is 0.02, which although low in absolute terms is relatively much higher than the one-step probability of 0.004. The system remains in a moderate stress regime after one year with a probability of approximately 0.70, and transitions to a calm state with a probability of 0.27, meaning complete recovery is much more likely than tipping back into acute crisis. From a

[16]The stationary distribution need not match empirical state shares. At a basic level, the stationary distribution is a model-implied long-run probability distribution, whereas the empirical share is a finite sample average of realized states. Even if the Markov chain is correct, the former converges to the latter only *asymptotically*. In a finite sample, sampling variation can easily produce differences. In this case, transitions from moderate to calm are more likely than calm to moderate ($\hat{p}_{10} > \hat{p}_{01}$), which shifts probability mass toward calm in the stationary distribution relative to the empirical frequencies. Estimation and sampling error is another source, because it gets magnified by a nonlinear transformation of the estimated transition probabilities.

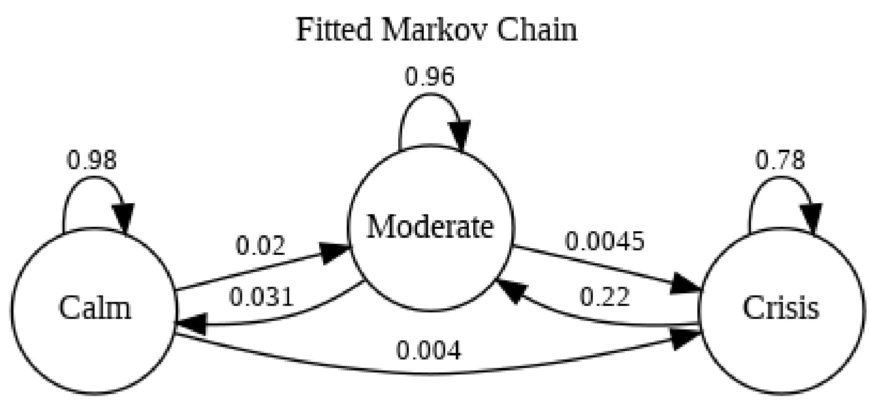
crisis state, the system moves to a moderate regime with probability 0.74 after a year and directly reverts to calm with probability 0.19 and remains in crisis with probability 0.06. Generally, the probability of remaining in a crisis regime falls drastically as h increases.

If the one-step matrix accurately captures the evolution of the system, there should not be any systematic differences between $\widehat{P}^h$ and the empirical multi-step matrix across horizons (observed differences reflect sampling variability). Table 9 summarizes the discrepancies across horizons.

Table 9: Multi-Step Transition Fit Summary

Horizon h (months)	Mean Abs. Diff.	Max Abs. Diff.
2	0.0151	0.0574
3	0.0161	0.0476
6	0.0511	0.1899
12	0.0380	0.1458

At $h = 2$ and $h = 3$, the mean absolute differences are 0.015 and 0.016, respectively, and the maximum absolute difference is below 0.06. Both maximums are from the low-count crisis regimes, thus I conclude the one-step matrix accurately captures the system's short-run evolution. Discrepancies do increase at $h = 6$ and $h = 12$, but again the largest deviations are for transitions involving the rare crisis regime. For the dominant calm and moderate stress regimes, model-implied multi-step probabilities are very close to empirical estimates. At $h = 12$, for example, the probability of remaining in the calm state is 0.7803 under the model, compared to 0.7890 empirically. The absolute differences are at worst on the order of 0.01, with several less than 0.01, and can be explained by sampling variation alone. I proceed with diagnostics for the first-order Markov approximation.



7 Model Diagnostics

There are several limitations of my approach. First, I assign regime labels based on the regime characteristics and my interpretation of their economic meaning. Although PCA and k-means identify clusters objectively, interpreting them is naturally subjective. Second, transition dynamics are contingent on the state space, and the state space is contingent on a host of data preprocessing steps and modeling choices. Third, a Markov chain is a descriptive model of transitions. It states *when* transitions may occur and with what probability, but unlike a structural model that explains *why* transitions occur, it is blind to economic reasoning. Fourth, all model-derived quantities associated with the crisis state are imprecise due to scarcity of crisis periods in the sample.

The first and third issues are not unique to this framework. Latent state-space models also rely on interpretation to assign economic meaning to inferred regimes, whereas my approach makes this step explicit. Moreover, such models are typically reduced-form and do not, by themselves, impose structural economic mechanisms on transition dynamics. To address the second and fourth issues, I conduct robustness checks for both the state space and the estimated transition dynamics. I also evaluate the adequacy of the Markov chain approximation by assessing the empirical plausibility of the Markov property and time-homogeneity assumption. While not exhaustive, these diagnostics provide evidence on whether the baseline model offers a reasonable approximation to macro-financial regime dynamics.

7.1 First-Order Markov Adequacy

The Markov property states that conditional on the current state, the past contains no additional information about the next transition, meaning

$$\mathbb{P}(X_{t+1} = j \mid X_t = i, X_{t-1} = i_{t-1}, X_{t-2} = i_{t-2}, \dots) = \mathbb{P}(X_{t+1} = j \mid X_t = i).$$

I assess this restriction by comparing the estimated first-order transition probabilities with empirical second-order transition probabilities[17]. As stated before, I estimate first-order probabilities using

$$\hat{p}(i, j) = \frac{\sum_{t=1}^{n-1} \mathbf{1}\{x_t = i,\, x_{t+1} = j\}}{\sum_{t=1}^{n-1} \mathbf{1}\{x_t = i\}}.$$

The second-order probabilities condition on the current and previous state,

$$\hat{p}^{(2)}(k, i, j) = \widehat{\mathbb{P}}(X_{t+1} = j \mid X_t = i, X_{t-1} = k) = \frac{\sum_{t=2}^{n-1} \mathbf{1}\{x_{t-1} = k,\, x_t = i,\, x_{t+1} = j\}}{\sum_{t=2}^{n-1} \mathbf{1}\{x_{t-1} = k,\, x_t = i\}}.$$

If the Markov property holds, then the first and second-order probabilities should be equal, and conditioning on a lagged state k should not change the probability distribution of X_{t+1} conditional

[17]The second-order comparison tests whether additional lagged state information improves one-step prediction and is distinct from the multi-step transition analysis, which tests whether the estimated one-step matrix reproduces observed dynamics over longer horizons.

on current state i [Wilks, 2006]. Table 10 shows the differences between estimated second and first-order transition probabilities for two k and i pairs with a meaningful number of observations.

Table 10: Second-Order vs. First-Order Transition Differences

(x_{t-1}, x_t)	Count	Mean Abs. Diff.	Max Abs. Diff.	$\Delta\hat{P}(\cdot \to 0)$	$\Delta\hat{P}(\cdot \to 1)$
$(0, 0)$	241	0.0024	0.0036	0.0034	-0.0036
$(1, 1)$	215	0.0023	0.0035	-0.0035	0.0033

These pairs (calm and calm, moderate and moderate) are the two most prevalent second-order histories. For both, the maximum absolute difference is at most 0.0036. Conditioning on X_{t-1} adds effectively no predictive power over merely conditioning on X_t, indicating the first-order Markov approximation is at least plausible. For a formal comparison, I use a likelihood ratio test. Following [Wilks, 2006], let N_{kij} denote the number of observed triples in the fitted regime sequence with $(x_{t-1}, x_t, x_{t+1}) = (k, i, j)$. The first-order log likelihood is

$$\log L_1 = \sum_{k,i,j} N_{kij} \log \hat{p}(i, j),$$

while the second-order log likelihood is

$$\log L_2 = \sum_{k,i,j} N_{kij} \log \hat{p}^{(2)}(k, i, j).$$

(k, i, j) pairs with $N_{kij} = 0$ drop out of both sums. I test the null hypothesis

$$H_0 : \mathbb{P}(X_{t+1} = j \mid X_t = i, X_{t-1} = k) = \mathbb{P}(X_{t+1} = j \mid X_t = i) \quad \text{for all } i, j, k \in \mathcal{S},$$

against the alternative that transition probabilities depend on both X_t and X_{t-1}, using the likelihood-ratio statistic

$$LR = 2(\log L_2 - \log L_1).$$

Under the null, the test statistic is asymptotically chi-square,

$$LR \xrightarrow{d} \chi^2_{s(s-1)^2}.$$

With $s = 3$ states, there are $s(s-1)^2 = 3(2)^2 = 12$ degrees of freedom. The test gives

$$\log L_1 = -69.7571,$$
$$\log L_2 = -67.6140,$$
$$LR = 4.2862,$$

with a p-value of 0.9777. I thus fail to reject the first-order Markov restriction[18].

7.2 Time Homogeneity

A DTMC assumes time homogeneity, meaning that transition probabilities are constant over time. This is a strong assumption for a model of macro-financial regimes, but one that is necessary because it makes the exercise empirically tractable. To test the plausibility of this assumption, I retain the full-sample discrete process X_t and try three approaches for estimating transition dynamics[19]. First, I estimate the transition probabilities in separate subperiods. Second, I compute rolling transition probabilities, using a sliding window of 10 years. Lastly, I augment the empirical frequency estimator by conditioning on covariates.

For subperiod analysis, I partition the sample into four periods corresponding to well-documented shifts in macro-financial conditions and with comparable sample sizes: 1986–1999 (post-stagflation expansion era), 2000–2009 (dot-com cycle and Global Financial Crisis), 2010–2019 (post-GFC recovery and low-volatility, rate, and growth regime), and 2020–2026 (Covid and subsequent inflation and tightening cycle). The periods do not overlap and cumulatively cover the entire sample. I then estimate transition matrices separately within each period and compare them to the full period estimate. See Appendix I for complete results.

I perform a likelihood ratio test again to formally test for time-varying transition probabilities. Again following [Wilks, 2006], for R subperiods, the test statistic is

$$LR = 2\sum_{r=1}^{R}\sum_{i,j\in\mathcal{S}} N_{ij}^{(r)} \log\left(\frac{\hat{p}^{(r)}(i,j)}{\hat{p}(i,j)}\right),$$

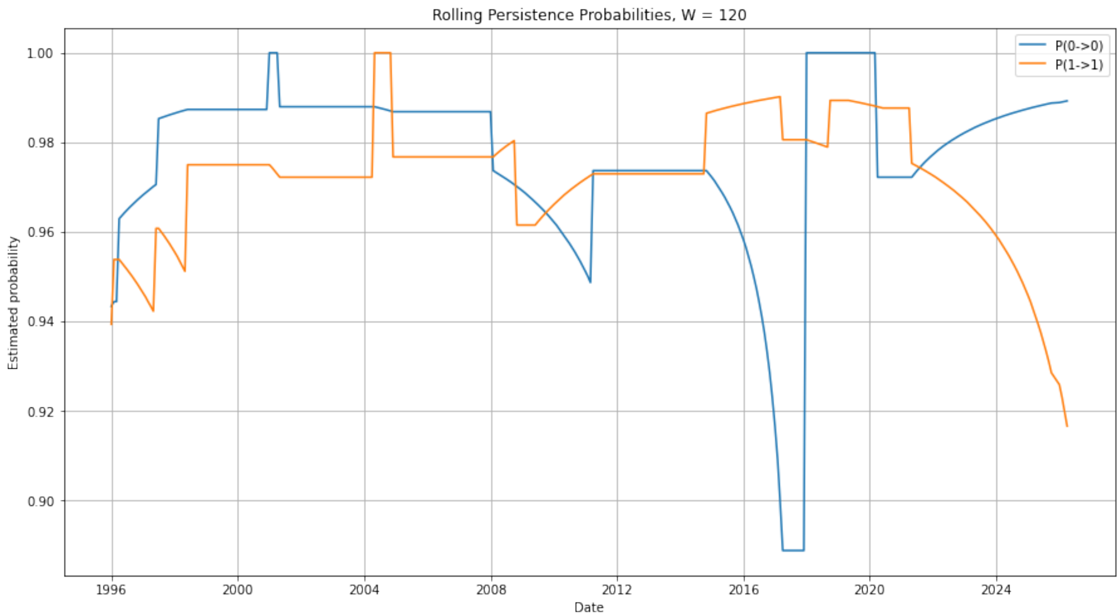
where $\hat{p}^{(r)}(i,j)$ is the estimated transition probability in subperiod r and $\hat{p}_{ij}$ is the pooled estimate. Under the null hypothesis of time homogeneity, the statistic is asymptotically χ^2 with $(R-1)s(s-1)$ degrees of freedom ($R = 4$ subperiods, $s = 3$ states). With 18 degrees of freedom, the test yields a statistic $LR = 15.55$ and a $p-$value of 0.624, thus I fail to reject the null hypothesis of time-invariant transition probabilities. The test confirms that the estimated transition structure is stable despite the substantial differences in macro-financial environments across the subperiods.

Rolling window estimation is a nonparametric approach to estimating time-varying parameters. For a window of size W and n total observations, the procedure first fits the model to observations $1, \ldots, W$, then $2, \ldots, W + 1$, and so on, until the final fit on observations $n - W + 1, \ldots, n$. By incrementally dropping the oldest observation and adding the newest observation, the procedure generates time-varying parameter estimates. A larger W results in smoother but fewer estimates, whereas a smaller W constrains the sample size for each fit and thus statistical power. Using a

[18]The test compares the restricted model (first-order Markov) to an unrestricted model (second-order Markov). The distribution of the test statistic is asymptotically chi-square, thus the low sample counts for some second-order transitions may make the test too conservative despite the test statistic of 4.29, which would otherwise constitute decisive non-rejection.

[19]Time homogeneity pertains to the probability of moving between states, not whether the state space is constant over time. Without a fixed state space and thus static labels, the transition probabilities cannot be compared.

baseline of $W = 0.25n$, for $n = 481$, I set $W = 120$, corresponding to a 10-year window. I estimate transition probabilities within each window using the empirical frequency estimator. In Figure 4, I plot the diagonal elements $\hat{p}_{00,t}$ and $\hat{p}_{11,t}$ over time (t marks the end of the 10-year window).



Despite the apparent time variation, the range for each estimated probability is actually quite narrow. The calm state's transition probability is bounded between 0.89 and 1.00, and the moderate stress states is between 0.92 and 1.00. The probabilities vary the most during periods of elevated financial stress (e.g., the GFC and Covid), but the overall level of persistence is high throughout the sample and indicates that time homogeneity is at least a plausible assumption.

7.3 Covariate-Conditioned Transitions

If the discrete state X_t is a sufficient statistic for all relevant information on macro-financial conditions, conditioning on external covariates (lagged or contemporaneous) should not affect transition probabilities [Filardo, 1998]. As a last robustness check, I test whether the 10Y–3M term spread, $s_t^{10y,3m}$, and the BAA–10Y credit spread, $c_t^{baa,10y}$ alter transition behavior *post* state space construction. Let W_t be a conditioning variable. I split the sample into two bins using its sample median. For bin $b \in \{\text{low}, \text{high}\}$, let

$$N_{ij}^{(b)} = \sum_{t=1}^{n-1} \mathbf{1}\{x_t = i,\, x_{t+1} = j,\, W_t \in B_b\},$$

where B_b denotes high or low elements of W_t. I estimate

$$\hat{p}^{(b)}(i,j) = \widehat{\mathbb{P}}(X_{t+1} = j \mid X_t = i, W_t \in B_b) = \frac{N_{ij}^{(b)}}{\sum_{\ell \in \mathcal{S}} N_{i\ell}^{(b)}}.$$

I then compare these covariate-conditioned probabilities to the baseline first-order transition probabilities $\hat{p}_{ij}$[20]. See Appendix J for full covariate-conditioned transition matrices.

Table 11: Covariate-Conditioned Transition Probability Deviations

Covariate	Bin	State i	N	Mean $\lvert\Delta\rvert$	Max $\lvert\Delta\rvert$	Largest Shift
$c_t^{baa,10y}$	Low	1	48	0.0485	0.0728	$\Delta p_{10} = 0.0728$
	High	0	52	0.0378	0.0568	$\Delta p_{01} = 0.0568$
	High	1	175	0.0133	0.0200	$\Delta p_{10} = -0.0200$
	Low	0	196	0.0100	0.0151	$\Delta p_{01} = -0.0151$
$s_t^{10y,3m}$	High	0	37	0.0046	0.0069	$\Delta p_{01} = 0.0069$
	Low	1	29	0.0030	0.0045	$\Delta p_{12} = -0.0045$
	Low	0	211	0.0008	0.0012	$\Delta p_{01} = -0.0012$
	High	1	194	0.0004	0.0007	$\Delta p_{12} = 0.0007$

Results for pairs with ≥ 20 observations. $\Delta p(i,j) = \hat{p}^{(b)}(i,j) - \hat{p}(i,j)$ is difference between covariate-conditioned transition probability and baseline first-order transition probability. Bin on covariate median.

Table 11 reports results per combination of covariate, bin, and current state i for pairs where there are at least 20 observations. Conditioning on the term spread has next to no effect for all bins and states. The largest absolute difference is 0.0069, for the transition from calm to moderate stress. The results for conditioning on the investment grade credit spread are more revealing. When credit spreads are low, and the system is in a moderate stress regime, conditioning raises the probability of transitioning to calm by 0.0728. When the system is in a calm regime and credit spreads are high, conditioning raises the probability of going from calm to stressed by 0.057. These results suggest credit conditions meaningfully signal deterioration and normalization, and that the state construction procedure partially masks that signal. Nonetheless, the magnitudes of the deviations are modest, and conditioning generally preserves the transition structure. The spread variables provide limited explanatory power and the first-order Markov approximation is reasonable[21].

[20] I already used both credit and term spreads as features in PCA to construct the state space. Conditioning on both doesn't explicitly test the Markov property, but whether the process of compressing signals using PCA and discretizing continuous features using k-means on the scores loses economically important signals. Equivalently, conditioning quantifies any incremental explanatory power they have.

[21] As an additional check, I considered conditioning on external stress indicators such as VIX and STLFSI. These variables introduce modest shifts in transition probabilities, suggesting that the discrete state representation may not fully capture the *intensity* of financial conditions. The effects are small and do not alter the main conclusions, so I focus on the baseline specification.

8 Conclusion

Latent state approaches to modeling macro-financial regimes specify a full probabilistic model and require jointly estimating the state space and transition dynamics. As an alternative, I implement a reduced-form approach that separates the process of state space construction from estimating transition dynamics. I construct a deterministic mapping from macro-financial time series to a discrete stochastic process by first applying principal component analysis to isolate changing financial conditions, and second partitioning that transformed feature space into clusters which I define as regimes. Having characterized regimes, I estimate transition probabilities using empirical frequencies and compute transition dynamics, such as multi-step transition matrices, the stationary distribution, recurrence times, and expected hitting times.

Applying my methodology to a monthly panel of U.S. macro-financial data spanning January 1986–March 2026, I identify a system with three regimes, which based on the historical record I interpret as calm, moderate-stress, and crisis. The calm regime accounts for approximately 52% of observations in the sample, and exhibits high persistence ($\hat{p}_{00} \approx 0.98$). The expected duration is 3 years, and the stationary probability is 0.55. The moderate-stress regime comprises roughly 46% of the sample, and is highly persistent ($\hat{p}_{11} \approx 0.96$) with an expected duration of 2 years. The regime appears pre-crisis, indicating that the deterioration of financial conditions makes markets vulnerable to shocks, and is more prevalent post-crisis, which suggests that extreme financial conditions during a crisis normalize gradually. The crisis regime is rare, representing less than 2% of observations in the sample, and aligns with known events such as Covid and the GFC. It has an expected duration of 4–5 months, a low stationary probability of 0.02, and large expected hitting times from non-crisis states. The estimated transition matrix shows that the system primarily moves between adjacent states, such as calm to moderate, and moderate to crisis, and only very rarely moves directly from calm to crisis. The multi-step transition probabilities show a similar phenomenon, albeit with slower diffusion over longer horizons. For each metric, the model-implied measure is close to the empirical analogue, with small discrepancies for the crisis state owing to how infrequent it is in the sample.

I perform several robustness checks. First, I test alternative cluster specifications ($K = 2$ and $K = 4$) and find they have modestly better clustering metrics but considerably worse economic interpretability and transition dynamics. Second, I compare first and second-order empirical transition probabilities, and complement the test with covariate-dependent transitions. Transition dynamics are consistent across all specifications, thus I conclude the first-order Markov approximation is reasonable. Third, I assess the assumption of time-homogeneity using subperiods and rolling windows, and again find consistent transition dynamics.

The simplicity of my approach comes at a cost. Because I construct the state space first and then treat it as fixed, I do not account for uncertainty or error in state assignment when estimating transition probabilities. Moreover, modeling choices such as the number of principal components, number of clusters, and beginning or end date of the sample, materially impact the state space but are considered in isolation, without regard to the resulting transition dynamics or overall quality of the model's fit. A full probabilistic model avoids this issue. Possible extensions to my approach include time-varying transition probabilities, which can be implemented nonparametrically using subperiods or rolling windows, clustering methods that account for temporal dependence, and representing uncertainty around regime assignments by computing distributions of transition matrices.

Ultimately, the results show that a simple approach can recover a coherent and economically meaningful description of macro-financial regimes.

A Feature Transformations

For equity market conditions, I use monthly S&P 500 returns computed as

$$r_t^{\mathrm{SP500}} = 100 \times \left(\frac{P_t}{P_{t-1}} - 1 \right),$$

where P_t denotes the end-of-month index level. For macroeconomic variables such as CPI and industrial production, I compute year-over-year growth rates

$$\pi_t = 100 \left(\frac{\mathrm{CPI}_t}{\mathrm{CPI}_{t-12}} - 1 \right), \qquad g_t^{ip} = 100 \left(\frac{\mathrm{IP}_t}{\mathrm{IP}_{t-12}} - 1 \right).$$

where CPI_t and IP_t are the levels of the series in month t. This transformation removes low-frequency trends and mitigates seasonality. For changes in the condition of the labor market, I first difference the reported monthly civilian unemployment rate (measured in percent of the labor force)

$$\Delta u_t = u_t - u_{t-1}.$$

I also construct term spreads using the 3-month, 2-year, and 10-year Treasury yields,

$$\begin{aligned} s_t^{10y,3m} &= y_t^{10y} - y_t^{3m}, \quad & s_t^{10y,2y} &= y_t^{10y} - y_t^{2y} \\ \Delta s_t^{10y,3m} &= s_t^{10y,3m} - s_{t-1}^{10y,3m}, \quad & \Delta s_t^{10y,2y} &= s_t^{10y,2y} - s_{t-1}^{10y,2y}, \end{aligned}$$

and credit spreads using AAA and investment grade bond yields and the 10-year Treasury yield,

$$\begin{aligned} c_t^{baa,aaa} &= y_t^{baa} - y_t^{aaa}, \quad & c_t^{baa,10y} &= y_t^{baa} - y_t^{10y} \\ \Delta c_t^{baa,aaa} &= c_t^{baa,aaa} - c_{t-1}^{baa,aaa}, \quad & \Delta c_t^{baa,10y} &= c_t^{baa,10y} - c_{t-1}^{baa,10y}, \end{aligned}$$

to capture the slope of the yield curve and the pricing of credit risk[22]. Spreads are in levels, and thus indicate the current state of macro-financial conditions. This is especially true for credit spreads. For example, an 8% spread suggests much higher credit risk than a 3% spread. The change in spread is more akin to a flow variable, as it indicates whether conditions are improving or deteriorating. I thus include first-differenced spreads, which have the added benefit of removing linear time trends that may otherwise dominate a model of co-movement between financial variables[23].

[22]The difference in yields reflects compensation investors require for holding bonds outside their preferred investment horizon (longer maturity bonds have greater interest rate risk, are more susceptible to variable inflation eroding the purchasing power of their fixed cashflows, and are less liquid) and for bearing credit risk (all else equal, investors will pay less for a bond with risky cashflows).

[23]It may appear redundant or excessive to include spreads in levels and first-differences. My rationale is that doing so captures both regime characteristics (levels), which is economically meaningful long-run information, and higher frequency adjustments to the regime (first-differences). For example, during a stress period, credit spreads may be elevated. A levels only variable wouldn't distinguish between 7% and 8% spreads, as both signal distressed credit markets, but the change variable captures rapid repricing reflected in further relative degradation of 1 percentage point, which is over a 12.5% increase from 7%.

B Time-Series Diagnostics

I assess the time-series properties of the set of 12 transformed variables used in principal component analysis, including equity returns, inflation, industrial production growth, change in the unemployment rate, and credit and term spreads in levels and first-differences. PCA assumes that the input data summarizes a stable covariance structure, and thus the presence of serial correlation, conditional heteroskedasticity, and structural breaks can uniquely undermine that assumption. Some serial correlation is to be expected in macro-financial time series, but high serial correlation indicates that each individual observation in a time series contributes less unique information about changing conditions and therefore the aggregate amount of information conveyed is less than the sample size would suggest. To test for serial correlation, I use a Ljung-Box test with 6, 12, and 24 lags. The chosen lags reflect a range, bounded below by the medium horizon at 6 months and above by the length of a business cycle at 2 years. Conditional heteroskedasticity means that a time series' volatility is time-varying. For example, equity returns are much more volatile in stress periods than calm periods. Series that exhibit conditional heteroskedasticity can dominate estimated principal components because PCA finds directions in the feature space that explain the most variance, and thus such series will mechanically have higher loadings. This is useful for identifying stress regimes, but not for balanced macroeconomic regimes. I use Engle's ARCH Lagrange multiplier test with 12 lags. The most serious challenge to PCA is the presence of a structural break, which means the relationship between variables fundamentally changes over time, and thus the covariance structure is time-varying. If a break is present, sub-sample PCA, splitting at the break point, is more defensible than full-sample PCA. I use a CUSUM test based on residuals from a simple AR(1) regression with an intercept as a stability diagnostic.

Table 12: Time-Series Diagnostics for PCA Features

Variable	ADF p	KPSS p	Classification	LB(6) p	LB(12) p	LB(24) p	ARCH LM p	CUSUM p
r_t^{SP500}	0.0000	0.1000	Stationary	0.5446	0.4566	0.5259	0.0053	0.5589
π_t (CPI)	0.0080	0.1000	Stationary	0.0000	0.0000	0.0000	0.0000	0.7243
g_t^{ip}	0.0036	0.0367	Inconclusive	0.0000	0.0000	0.0000	0.0000	0.5850
Δu_t	0.0000	0.1000	Stationary	0.0547	0.3985	0.9443	0.9998	0.4082
$s_t^{10y,3m}$	0.0018	0.0217	Inconclusive	0.0000	0.0000	0.0000	0.0000	0.6347
$s_t^{10y,2y}$	0.0047	0.1000	Stationary	0.0000	0.0000	0.0000	0.0000	0.5340
$c_t^{baa,aaa}$	0.0000	0.1000	Stationary	0.0000	0.0000	0.0000	0.0000	0.3051
$c_t^{baa,10y}$	0.0040	0.0581	Stationary	0.0000	0.0000	0.0000	0.0000	0.6809
$\Delta s_t^{10y,3m}$	0.0000	0.1000	Stationary	0.0000	0.0000	0.0000	0.0115	0.9994
$\Delta s_t^{10y,2y}$	0.0000	0.1000	Stationary	0.0000	0.0000	0.0000	0.0028	0.9576
$\Delta c_t^{baa,aaa}$	0.0000	0.1000	Stationary	0.0000	0.0000	0.0000	0.0000	0.5835
$\Delta c_t^{baa,10y}$	0.0000	0.1000	Stationary	0.0000	0.0000	0.0000	0.0000	0.6080

Table 12 reports p values for each test, which I perform at an $\alpha = 5\%$ level. All variables exhibit serial correlation at each lag except for equity returns and the change in the unemployment rate.

For efficient markets, the absence of serial correlation and thus predictability in broad index returns is to be expected. Similarly, the unemployment rate in levels is serially correlated but the *change* month to month resembles white noise. The serial correlation in inflation, industrial production, and credit and term spreads indicates they contain slow-moving information about regimes, and that serial correlation in derived principal component scores may reflect serial correlation in the underlying variables and thus explains regime persistence. Economically, this is sensible. ARCH LM p-values are nearly zero for all variables except Δu_t, which has a p-value of nearly 1. The unemployment rate in levels is a low-frequency variable, and large changes are attributable to discrete events like recessions followed by slow normalization. Every other variable has statistically significant conditional heteroskedasticity. Again, this is to be expected because these variables empirically exhibit volatility clustering, especially during crises. PCA thus will be sensitive to crisis periods, and I expect the clustering algorithm to identify a crisis or acute stress regime as a result. Again, this is fine economically because volatility clustering is a hallmark of financial stress periods. Most importantly, CUSUM p-values are all comfortably above 5%. The lowest is the Baa-Aaa credit spread, at 0.31. There is no evidence of a structural break in any series based on this test. Based on the CUSUM test results and mostly stationary series, the time series are stable enough and I proceed with PCA over the full sample.

C Feature Variable Correlation

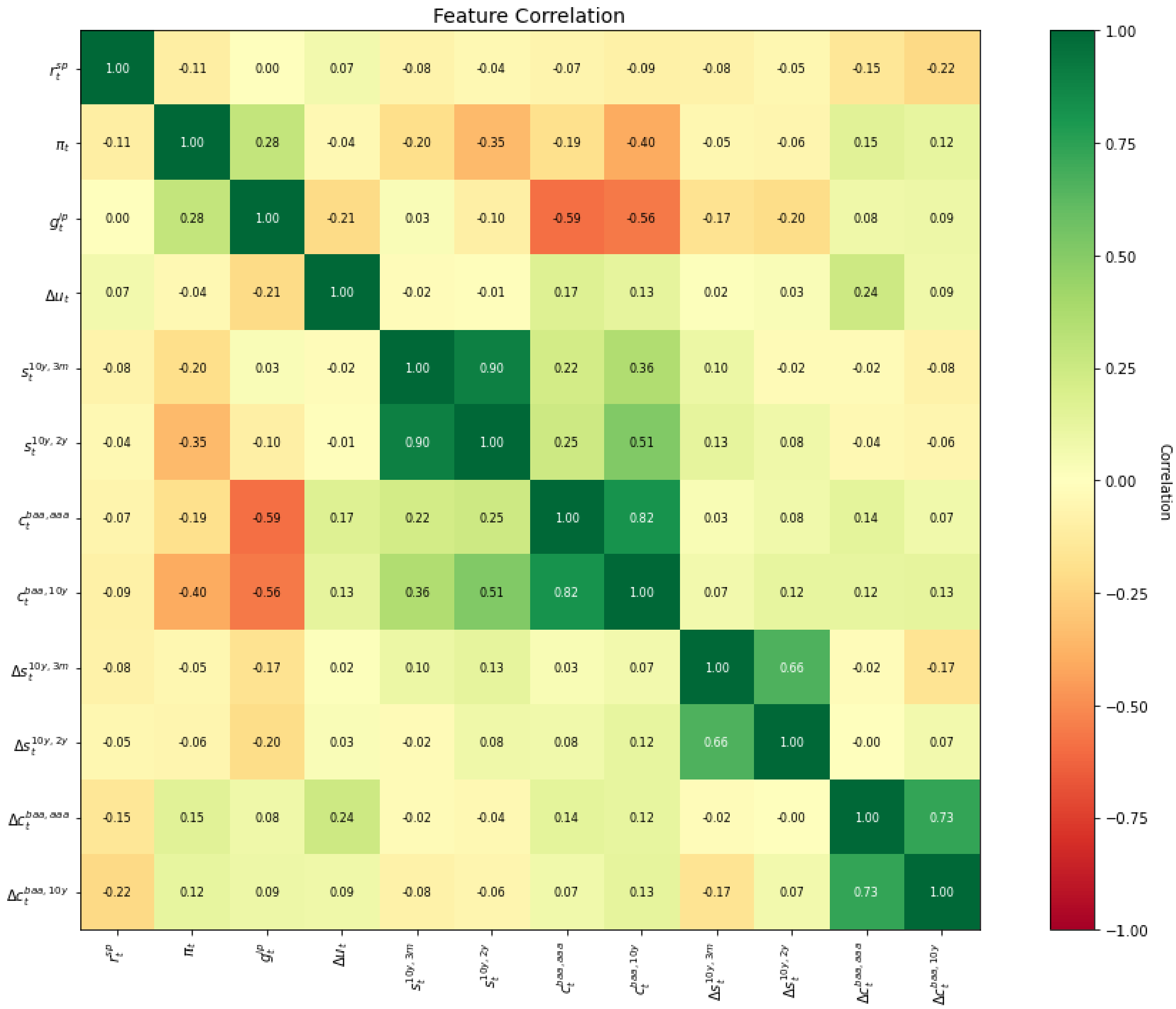
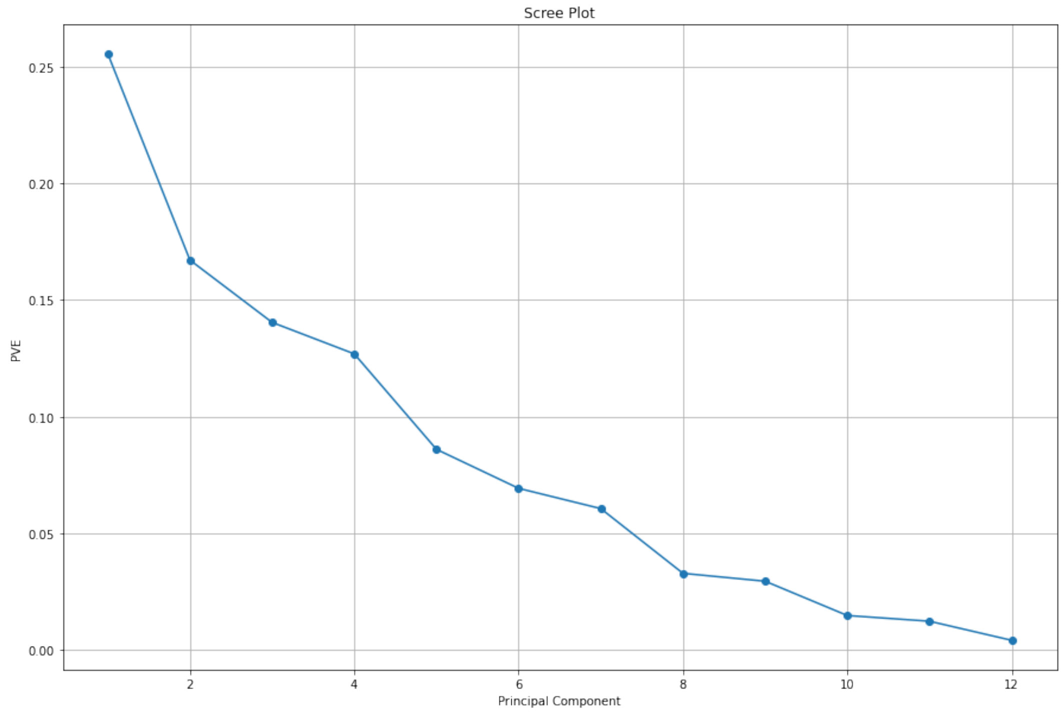


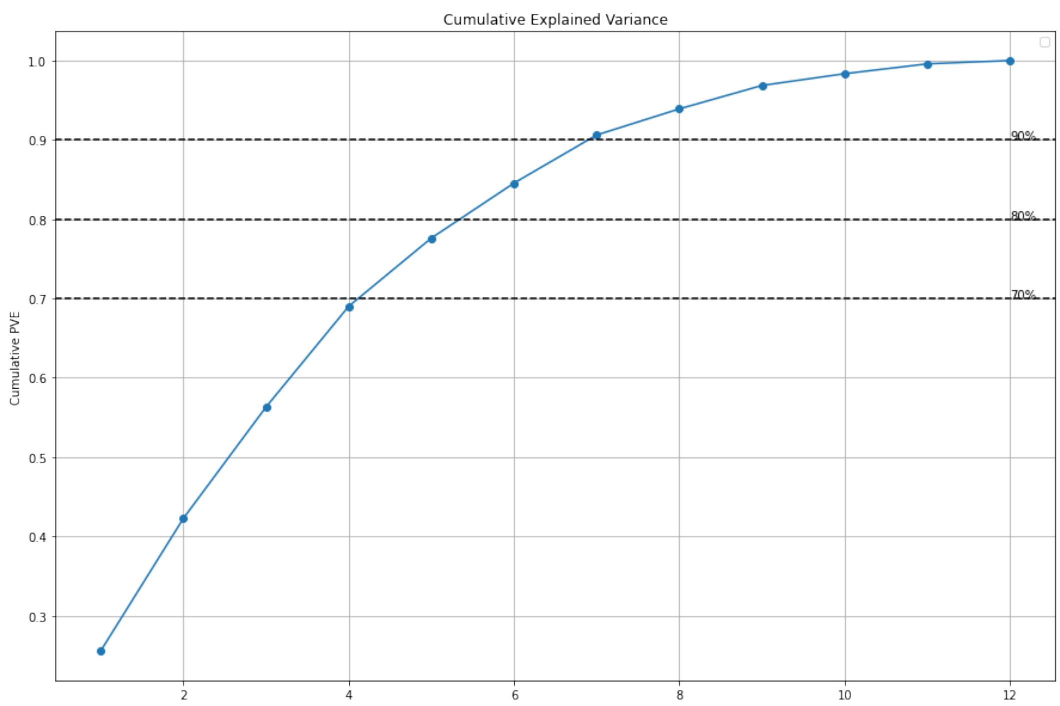
Figure 5 shows a correlation matrix for PCA variables, using the complete sample (Jan. 1986 - Mar. 2026). Credit and term spreads are highly correlated. The term spreads exhibit a correlation of 0.90, because they both measure the slope of the yield curve but use different short-term anchors (2Y vs. 3M). The Baa-Aaa and Baa-10Y spreads have a correlation of 0.82, again because both measure credit risk premia but with different benchmarks (AAA vs. 10Y). The same pattern appears for first-differenced spreads (0.66 and 0.73, respectively). Credit spreads and industrial production growth are strongly negatively correlated, at -0.59 and -0.56, indicating that weak real macroeconomic activity is linked with financial stress proxied by wider credit spreads. Credit spreads in levels have modest positive correlation with term spreads (0.51 for 10Y-2Y and BAA-10Y, 0.36 for 10Y-3M and BAA-10Y), indicating that wider credit spreads coincide with steeper yield curves. The correlation isn't merely attributable to the shared 10Y yield because of its different position in the spreads.

During a recession or early post-recession recovery, distress and credit defaults rise while the yield curve steepens as short rates fall or markets anticipate monetary policy easing. Inflation is modestly negatively correlated with all term and credit spreads in levels. High inflation is associated with flatter yield curves and lower credit spreads. The economic interpretation is that late in a business cycle, economic activity and inflation is high but credit conditions are still benign and markets do not anticipate rising short rates. Equity returns have zero or modestly negative correlations with all other variables. The largest (in magnitude) are with the change in credit spreads at -0.22 and -0.15. Contemporaneous equity returns generally resemble white noise, but the link to wider credit spreads may stem from risk aversion. When investors are risk-averse, such as during periods of distress, they demand greater compensation to hold risky assets (e.g. equities, bonds with credit risk) over risk-free alternatives, and thus pay less (or equivalently, demand higher yields and expected returns). Finally, changes in the unemployment rate have minimal correlation with other variables. The notable correlations are with the growth in industrial production at -0.21, and the change in Baa-Aaa credit spreads at 0.24. Rising unemployment coincides with faltering real economic activity and weaker credit markets. The low correlations for equity returns and the change in the unemployment rate mean they may constitute unique signals of macro-financial conditions.

D PCA Results

Figure 6 shows percent of variance explained per principal component. Beyond the fifth principal component, the marginal increase in PVE is modest. Figure 7 shows cumulative percent of variance explained. The first four components capture nearly 70% of variation in the original dataset.





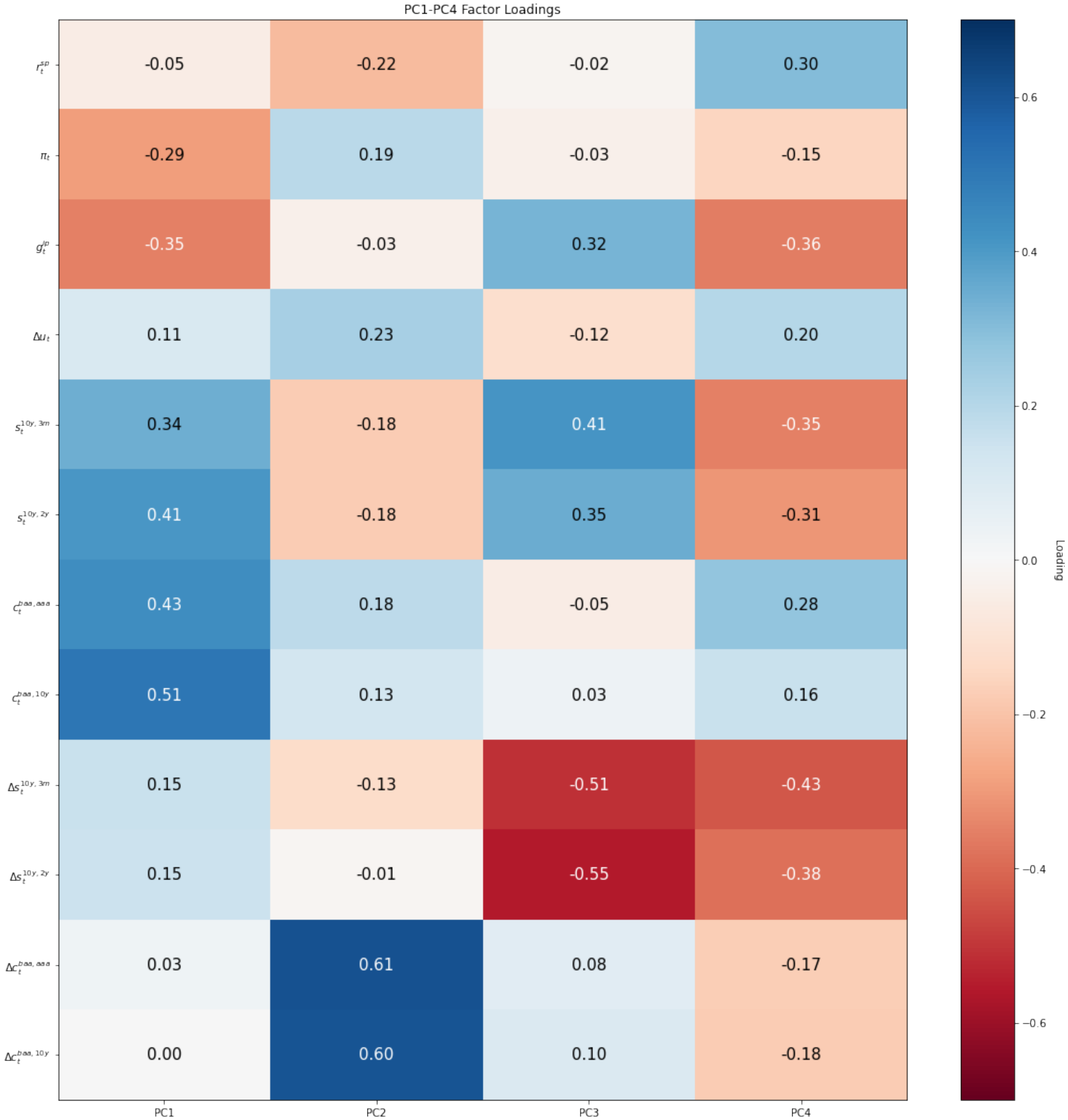


Figure 8 shows factor loadings for the first four principal components.

Table 13 reports complete loadings for each principal component. Loadings are eigenvectors from the spectral decomposition of the variance/covariance matrix.

Table 13: PCA Loadings

Variable	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9	PC10	PC11	PC12
r_t^{SP500}	-0.0496	-0.2200	-0.0208	0.3018	0.6333	-0.3655	-0.5605	0.0074	-0.0503	0.0709	0.0263	0.0207
π_t	-0.2948	0.1871	-0.0346	-0.1529	-0.1576	0.5602	-0.6611	-0.1745	-0.0840	0.1049	-0.1710	0.0046
g_t^{ip}	-0.3537	-0.0335	0.3227	-0.3577	0.1466	-0.0738	-0.0050	0.1844	0.7547	0.0824	-0.0399	-0.0521
Δu_t	0.1081	0.2327	-0.1197	0.1969	0.6230	0.5704	0.3391	-0.1225	0.1604	0.1072	0.0115	-0.0135
$s_t^{10y,3m}$	0.3401	-0.1752	0.4148	-0.3515	0.1302	0.1919	-0.1097	-0.0323	-0.1194	-0.1204	0.4173	0.5369
$s_t^{10y,2y}$	0.4058	-0.1802	0.3474	-0.3115	0.1410	0.0372	-0.0338	-0.1727	-0.1593	0.0259	-0.3129	-0.6395
$c_t^{baa,aaa}$	0.4335	0.1819	-0.0537	0.2757	-0.2143	0.0998	-0.3127	0.1729	0.3922	-0.1769	0.4627	-0.3425
$c_t^{baa,10y}$	0.5080	0.1282	0.0341	0.1551	-0.1418	-0.0857	-0.1112	0.0317	0.2762	0.3283	-0.5565	0.4070
$\Delta s_t^{10y,3m}$	0.1534	-0.1273	-0.5111	-0.4347	0.0909	0.0587	-0.0422	0.5465	-0.1107	0.4124	0.1077	-0.0631
$\Delta s_t^{10y,2y}$	0.1523	-0.0143	-0.5527	-0.3847	0.0861	-0.1704	-0.0637	-0.5101	0.2601	-0.3808	-0.0521	0.0833
$\Delta c_t^{baa,aaa}$	0.0274	0.6076	0.0772	-0.1733	0.2121	-0.1231	-0.0673	0.4314	-0.1893	-0.5006	-0.2349	0.0410
$\Delta c_t^{baa,10y}$	0.0017	0.6025	0.1019	-0.1762	0.0300	-0.3444	0.0030	-0.3373	-0.0981	0.4957	0.3210	-0.0625

Figure 9 shows time-series plots of PC1-PC4 scores, with NBER recessions overlayed.

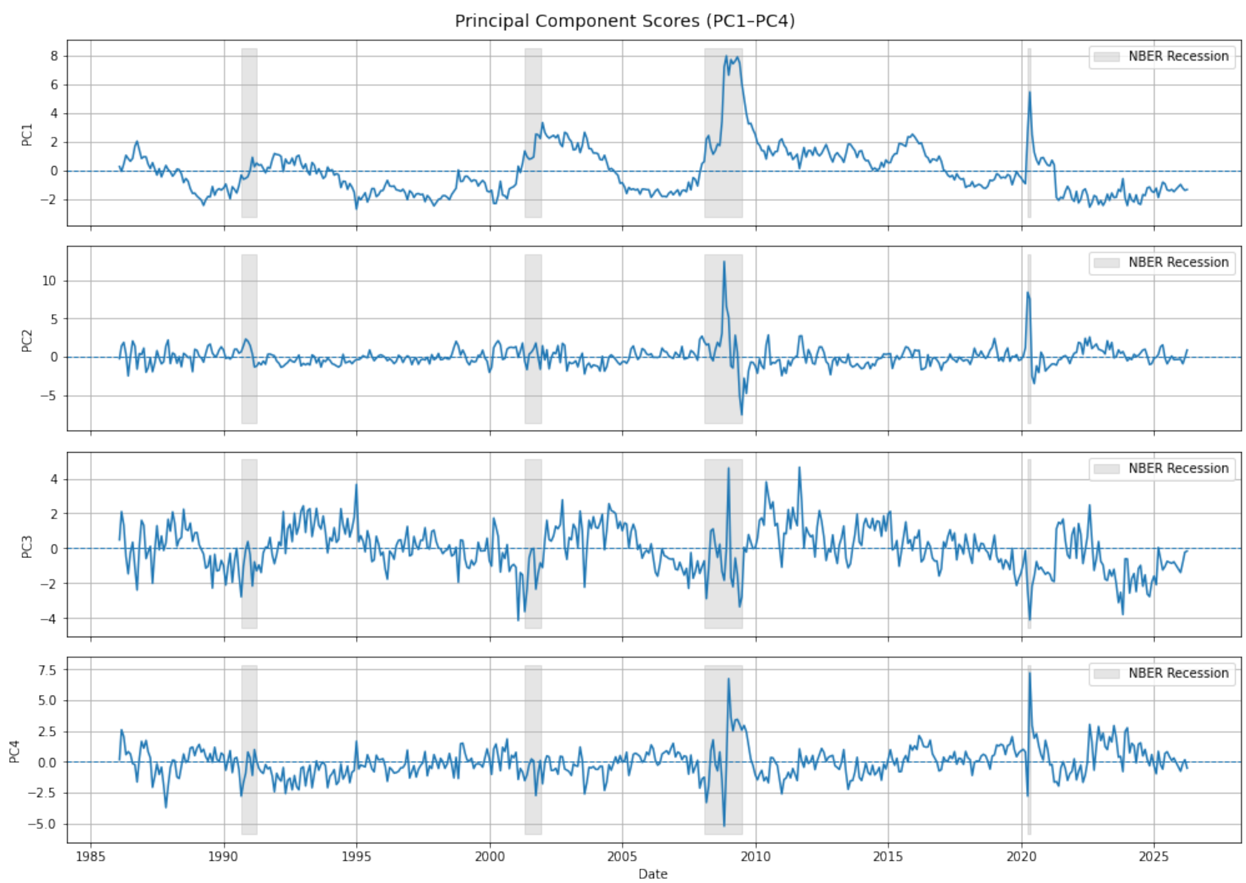


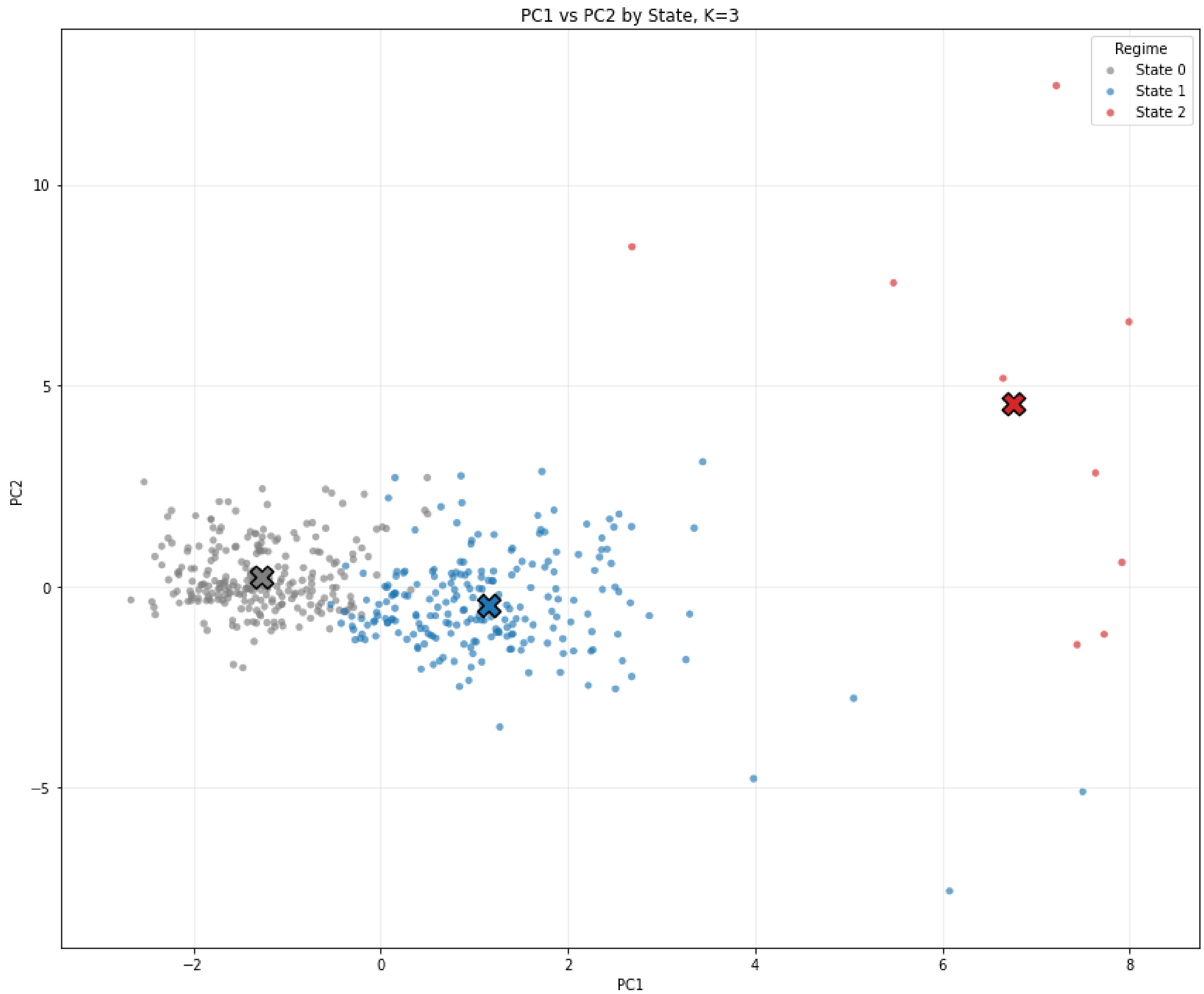
Table 14 shows the time-series properties of the first four principal components. PC4's stationarity testing results are mixed. All exhibit serial correlation and conditional heteroskedasticity. PC3 and PC4 exhibit evidence of a structural break, specifically parameter instability in the AR(1).

Table 14: Time-Series Diagnostics for Principal Component Scores

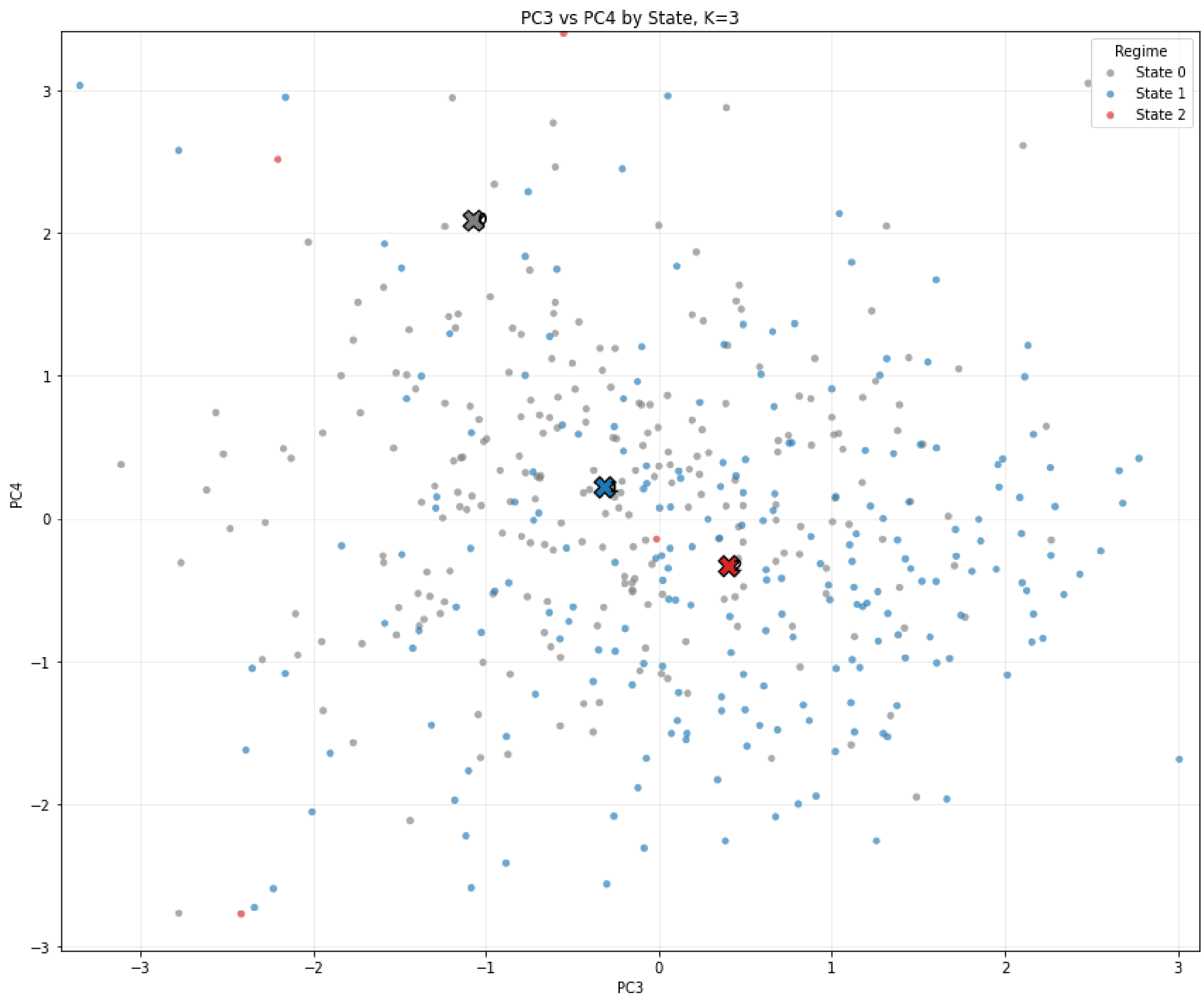
Series	ADF p	KPSS p	Classification	LB(6) p	LB(12) p	LB(24) p	ARCH LM p	CUSUM p
PC1	0.0112	0.1000	Stationary	0.0000	0.0000	0.0000	0.0000	0.4220
PC2	0.0000	0.1000	Stationary	0.0000	0.0000	0.0000	0.0000	0.8467
PC3	0.0002	0.0865	Stationary	0.0000	0.0000	0.0000	0.0000	0.0340
PC4	0.0000	0.0112	Inconclusive	0.0000	0.0000	0.0000	0.0000	0.0118

E Cluster Fits

Figure 10 and Figure 11 show a projection of the fitted clusters onto Principal Components 1 and 2, and separately onto 3 and 4. Each point is a monthly observation, defined by a vector of scores $(Z_{t1}, Z_{t2}, Z_{t3}, Z_{t4})$, and colored by state $X_t = k$ for $(Z_{t1}, Z_{t2}, Z_{t3}, Z_{t4}) \in C_k$. The markers denote cluster centroids.



The PC1-PC2 plane shows the clearest cluster separation. Observations in the calm regime, State 0, have mostly negative PC1 values, whereas observations in the moderate stress regime, State 1, have mostly positive PC1 values. The crisis regime, State 2, is scattered in the right corner of the plane, where its observations take extreme positive values for both PC1 and PC2. The geometry accords with my earlier interpretation. PC1 is a broad financial conditions factor that differentiates all regimes, whereas PC2 is a complementary stress factor that most clearly differentiates the crisis regime from the others. The clusters show much weaker separation in the PC3-PC4 plane. Although the centroids are differentiated, their observations are scattered and intermixed. The overall range for PC scores is tighter (-3 to 3 along both axes) than in the PC1-PC2 plane. These results are to be expected, because PC3 and PC4 explain considerably less variation than the first two components and I interpret them as second-order adjustment dynamics rather than first-order level effects. PC3 and PC4 are best thought of as contributing additional but more diffuse information about macro-financial conditions.



Overall, the clustering fit is reasonable for the purposes of regime construction. The fit in the PC1-PC2 plane is especially compelling, as it shows well-separated clusters and a right-tail crisis regime.

F Clustering Robustness Check

As a robustness check for the number of principal components and clusters, K, I compare cluster fits using PC1-PC4 and PC1-PC5 and $K = 2, 3, 4$. Comparing cluster fits requires external statistical criteria[24]. I use the silhouette score, the Calinski-Harabasz index, and the Davies–Bouldin index[25]. For the silhouette score and CH index, higher values are better. For the DB index, smaller values are better. As a tiebreaker and sanity check, I evaluate the economic properties of the resulting regimes and look for variation in cluster shares, average durations, and the one-month flip rate,

$$\frac{1}{n-1}\sum_{t=2}^{n}\mathbf{1}\{x_t \neq x_{t-1}\}.$$

The one-month flip rate is the empirical frequency of regime switching, and thus lower values indicate more persistent regimes. Table 15 reports results per K and set of principal components. The silhouette score is increasing in K. The CH and DB indices fall with higher K[26]. The statistical criteria are split, with $K = 3$ the middle result for all metrics, but the flip rate decisively favors $K = 3$. For both PC1-PC4 and PC1-PC5, $K = 3$ has the lowest flip rate of 0.0333. Economically, this is most desirable because regimes should not excessively switch month to month.

Table 15: Clustering Robustness Across K

Specification	K	Silhouette	Calinski–Harabasz	Davies–Bouldin	Flip Rate
PC1–PC4	2	0.2772	150.9750	1.4460	0.0396
PC1–PC4	3	0.2892	146.2461	1.2622	0.0333
PC1–PC4	4	0.2954	136.6958	1.1314	0.0417
PC1–PC5	2	0.2516	129.8163	1.5570	0.0396
PC1–PC5	3	0.2615	121.9714	1.3906	0.0333
PC1–PC5	4	0.2679	115.7344	1.2557	0.0417

Table 16 reports the cluster shares and average durations for the baseline PC1–PC4 specification. The two-state model produces two large and persistent states, but fails to isolate a distinct crisis regime. Given the weight of economic evidence for the existence of such a regime, this is un-

[24] The k-means objective function decreases as K increases because more clusters naturally reduce intra-cluster dispersion. Consider the extreme case of $K = n$, meaning one cluster per observation. The within-cluster sum of squares, and thus the objective function, is 0.

[25] The silhouette score measures how close observations are to their assigned cluster relative to neighboring clusters [scikit-learn developers, 2025c], whereas Calinski–Harabasz index compares intra-cluster dispersion to inter-cluster dispersion [scikit-learn developers, 2025a]. The Davies–Bouldin index compares intra-cluster dispersion to inter-cluster separation [scikit-learn developers, 2025b]. I implement each using standard libraries in scikit-learn.

[26] For the silhouette score, $K = 4$ is marginally better but the improvement from $K = 2$ to $K = 3$ is larger than from $K = 3$ to $K = 4$ (0.0120 vs. 0.0062 for PC1-PC4, 0.0099 vs. 0.0064 for PC1-PC5). The CH index favors $K = 2$, as it has the highest values, but the fall from $K = 3$ to $K = 4$ is much larger than $K = 2$ to $K = 3$ (-9.55 vs. -4.73 for PC1-PC4), so $K = 4$ is meaningfully worse than $K = 3$. The DB index again shows $K = 4$ is best but with diminishing gains as K increases.

acceptable. Exactly as expected, the three-state model produces two dominant states, with shares 52% and 46%, and a rare crisis state with share 2%. The average run lengths are 36 months for the calm state, 28 months for the moderate state, and 4.5 months for the crisis state. The four-state model retains the two dominant states but splits the tail of the distribution into two very small clusters with shares of 1.66% and 1.25% and average run lengths of 2.67 and 2 months, respectively. I interpret this as fragmentation of the crisis regime, rather than a substantive fourth state.

Table 16: Regime Shares and Durations, PC1–PC4

	Share				Duration (months)			
K	0	1	2	3	0	1	2	3
2	0.5281	0.4719	–	–	25.40	22.70	–	–
3	0.5177	0.4636	0.0187	–	35.57	27.88	4.50	–
4	0.5052	0.4657	0.0166	0.0125	34.71	28.00	2.67	2.00

The primary reason to prefer the three-state specification is interpretability in terms of external validation measures (VIX, STLFSI, NBER Recession Indicator). Table 17 reports validation summaries for the PC1–PC4 models. The three-state model produces a clean hierarchy. The calm regime has low financial stress and a recession share near zero, and the moderate regime has moderately higher stress and recession exposure. The crisis state has mean VIX of 49.84, mean STLFSI of 4.72, and the recession indicator equals one in all observations assigned to that state.

Table 17: Validation Statistics by Cluster: PC1–PC4

K	State	Mean VIX	Mean STLFSI	Mean USREC
2	0	18.20	-0.30	0.02
2	1	20.89	0.39	0.14
3	0	18.26	-0.30	0.02
3	1	19.48	0.15	0.10
3	2	49.84	4.72	1.00
4	0	18.37	-0.31	0.02
4	1	19.08	0.11	0.09
4	2	33.98	2.05	0.63
4	3	53.37	5.26	1.00

PC5 is somewhat of a non-factor. Excluding PC5 does not materially affect the clustering fit for $K = 3$, as the state shares, average duration, flip rate, and transition structure are effectively the same as with the PC1-PC4 baseline. There is some downside, as PC5 results in weaker clustering metrics, but no upside. I proceed with PC1-PC4 and $K = 3$ to construct the state space.

G Multi-Step Transition Probabilities

As a diagnostic for the first-order Markov specification, I compute and compare empirical multi-step transition probabilities with those implied by the estimated Markov chain. For a time-homogeneous Markov chain with transition matrix P, the h-step transition probability is

$$p^{(h)}(i,j) = \mathbb{P}(X_{t+h} = j \mid X_t = i) = P^h(i,j).$$

For estimated $\hat{P}$, the h-step transition matrix is $\hat{P}^h$, where the (i,j) entry is

$$\hat{p}^{(h)}(i,j) = \hat{P}^h(i,j)$$

and can be computed using straightforward matrix multiplication. The empirical h-step transition probabilities use counts of transitions from $x_t = i$ to $x_{t+h} = j$ and normalize row-wise,

$$\tilde{p}^{(h)}(i,j) = \frac{\sum_{t=1}^{n-h} \mathbf{1}\{x_t = i,\, x_{t+h} = j\}}{\sum_{t=1}^{n-h} \mathbf{1}\{x_t = i\}}.$$

If the one-step transition matrix provides an adequate summary of regime dynamics, then $\hat{P}^h$ should approximate the empirical h-step transition probabilities. I compute both at $h = 2, 3, 6, 12$, and record the absolute differences. Table 18 reports the full comparison.

Table 18: Empirical vs. Markov-Implied Multi-Step Transition Probabilities

Horizon	Source	$0 \to 0$	$0 \to 1$	$0 \to 2$	$1 \to 0$	$1 \to 1$	$1 \to 2$	$2 \to 0$	$2 \to 1$	$2 \to 2$
	Empirical	0.9555	0.0364	0.0081	0.0538	0.9372	0.0090	0.0000	0.4444	0.5556
$h = 2$	Markov	0.9528	0.0400	0.0072	0.0609	0.9312	0.0079	0.0070	0.3871	0.6059
	Abs. Diff.	0.0026	0.0036	0.0009	0.0071	0.0061	0.0010	0.0070	0.0574	0.0504
	Empirical	0.9390	0.0528	0.0081	0.0717	0.9148	0.0135	0.0000	0.5556	0.4444
$h = 3$	Markov	0.9310	0.0594	0.0096	0.0887	0.9008	0.0106	0.0190	0.5080	0.4730
	Abs. Diff.	0.0080	0.0065	0.0015	0.0169	0.0140	0.0029	0.0190	0.0476	0.0286
	Empirical	0.8889	0.1029	0.0082	0.1390	0.8341	0.0269	0.0000	0.8889	0.1111
$h = 6$	Markov	0.8723	0.1136	0.0141	0.1626	0.8220	0.0154	0.0717	0.6990	0.2293
	Abs. Diff.	0.0166	0.0108	0.0059	0.0236	0.0121	0.0115	0.0717	0.1899	0.1182
	Empirical	0.7890	0.1899	0.0211	0.2646	0.7175	0.0179	0.1111	0.8889	0.0000
$h = 12$	Markov	0.7803	0.2024	0.0173	0.2766	0.7049	0.0185	0.1926	0.7430	0.0644
	Abs. Diff.	0.0087	0.0125	0.0038	0.0120	0.0126	0.0006	0.0815	0.1458	0.0644

H Second-Order Transition Probabilities

I report and compare first and second-order transition probabilities to assess whether lagged states beyond X_t contain additional predictive information about X_{t+1}. Table 19 reports second-order probabilities of transitioning to state j from state i at time t and lagged state k at time $t-1$. Several rows contain probabilities of 0 and 1, reflecting sample event counts of 0, 1, or 2.

Table 19: Second-Order Transition Probabilities

Lag State k	Current State i	$j=0$	$j=1$	$j=2$
0	0	0.9793	0.0166	0.0041
0	1	0.0000	1.0000	0.0000
0	2	0.0000	0.0000	1.0000
1	0	0.8571	0.1429	0.0000
1	1	0.0279	0.9674	0.0047
1	2	0.0000	0.0000	1.0000
2	0	–	–	–
2	1	0.0000	1.0000	0.0000
2	2	0.0000	0.2857	0.7143

Table 20 provides the full comparison and reports the mean and maximum absolute differences in transition probabilities and deviations.

Table 20: Full Second-Order vs. First-Order Comparison

Lag State k	Current State i	Count	Mean Abs. Diff.	Max Abs. Diff.	$\Delta P(\cdot \to 0)$	$\Delta P(\cdot \to 1)$
0	0	241	0.0024	0.0036	0.0034	-0.0036
1	0	7	0.0818	0.1227	-0.1187	0.1227
0	1	5	0.0239	0.0359	-0.0314	0.0359
1	1	215	0.0023	0.0035	-0.0035	0.0033
2	1	2	0.0239	0.0359	-0.0314	0.0359
0	2	1	0.1481	0.2222	0.0000	-0.2222
1	2	1	0.1481	0.2222	0.0000	-0.2222
2	2	7	0.0423	0.0635	0.0000	0.0635

Again, apparent deviations in the small-count rows reflect rare conditioning events. Rows with hundreds of observations ($(0,0)$ and $(1,1)$) show differences below 0.004.

I Subperiod Transition Probabilities

As part of the time homogeneity assessment, I provide transition matrices per subperiod. Table 21 reports the time window, number of observations, event counts, and transition probabilities for each subperiod. Let n_r denote the number of observations in subperiod r, let $N_{ij}^{(r)}$ denote the number of observed transitions from state i to state j in subperiod r, and let $\hat{p}^{(r)}(i,j)$ denote the corresponding row-normalized transition probability. Table 21 reports subperiod transition probabilities.

Table 21: Subperiod Transition Counts and Probabilities

Subperiod	n_r	State i	$N_{i0}^{(r)}$	$N_{i1}^{(r)}$	$N_{i2}^{(r)}$	$\hat{p}^{(r)}(i,\cdot)$
1986–1999 Jan 1986 – Dec 1999	168	0	98	3	0	(0.9703, 0.0297, 0.0000)
		1	4	62	0	(0.0606, 0.9394, 0.0000)
2000–2009 Jan 2000 – Dec 2009	120	0	51	2	0	(0.9623, 0.0377, 0.0000)
		1	1	57	1	(0.0169, 0.9661, 0.0169)
		2	0	1	6	(0.0000, 0.1429, 0.8571)
2010–2019 Jan 2010 – Dec 2019	120	0	33	0	0	(1.0000, 0.0000, 0.0000)
		1	1	85	0	(0.0116, 0.9884, 0.0000)
2020–2026 Jan 2020 – Mar 2026	73	0	58	0	1	(0.9831, 0.0000, 0.0169)
		1	1	10	0	(0.0909, 0.9091, 0.0000)
		2	0	1	1	(0.0000, 0.5000, 0.5000)

The lone two crisis events, the GFC and Covid, fall into the 2000-2009 and 2020-2026 subperiods, and thus 1986-1999 and 2010-2019 periods do not have crisis-regime estimated probabilities. There are only 9 crisis months across the whole sample, with 2 during Covid and 7 during the GFC, thus full sample crisis dynamics are primarily driven by the GFC. Across all subperiods, the calm and moderate stress regimes exhibit higher persistence, with $\hat{p}_{00}$ and $\hat{p}_{11}$ above 0.90. The system rarely transitions from calm to moderate and moderate to calm, but in the 2020-2026 subperiod, $\hat{p}_{01}$ is 0 and $\hat{p}_{10}$ is 0.09. The post-GFC period of 2010-2019 is the most stable. The system begins in the dominant moderate stress regime and transitions just once to a calm regime, where it remains for the rest of the subperiod. Crisis transition probabilities are highest for the last subperiod because there are only two observations. Overall, the transition dynamics for the calm and moderate stress regimes are stable across subperiods because variation is bounded. The evidence of time inhomogeneity, namely time-varying crisis regime dynamics, mechanically reflects how rare crisis periods are in the whole sample.

J Covariate-Conditioned Transition Probabilities

Tables 22 and 23 show results binning on the 10Y–3M term spread, $s_t^{10y,3m}$, and the BAA–10Y credit spread, $c_t^{baa,10y}$. The low bin contains no observations in the crisis state.

Table 22: Transition Probabilities, Binned on Term Spread

State i	Bin b	$N_{i0}^{(b)}$	$N_{i1}^{(b)}$	$N_{i2}^{(b)}$	Total	$\hat{p}^{(b)}(i,\cdot)$
0	High	36	1	0	37	(0.9730, 0.0270, 0.0000)
0	Low	206	4	1	211	(0.9763, 0.0190, 0.0047)
1	High	6	187	1	194	(0.0309, 0.9639, 0.0052)
1	Low	1	28	0	29	(0.0345, 0.9655, 0.0000)
2	High	0	1	6	7	(0.0000, 0.1429, 0.8571)
2	Low	0	1	1	2	(0.0000, 0.5000, 0.5000)

Table 23: Transition Probabilities, Binned on Credit Spread

State i	Bin b	$N_{i0}^{(b)}$	$N_{i1}^{(b)}$	$N_{i2}^{(b)}$	Total	$\hat{p}^{(b)}(i,\cdot)$
0	High	48	4	0	52	(0.9231, 0.0769, 0.0000)
0	Low	194	1	1	196	(0.9898, 0.0051, 0.0051)
1	High	2	172	1	175	(0.0114, 0.9829, 0.0057)
1	Low	5	43	0	48	(0.1042, 0.8958, 0.0000)
2	High	0	2	7	9	(0.0000, 0.2222, 0.7778)
2	Low	0	0	0	0	

Table 24 shows the mean and maximum absolute difference between covariate conditioned transition probabilities and baseline probabilities per credit and term spread bin, where

$$\Delta p^{(b)}(i,j) = \hat{p}^{(b)}(i,j) - \hat{p}(i,j).$$

For states with a sufficient number of observations to produce reliable inferences, conditioning on the term spread produces almost no change in transition probabilities. Conditioning on the credit spread does produce larger differences, notably for transitions out of the calm regime under high credit spreads and transitions out of the moderate regime under low credit spreads. The signs are intuitive. Elevated credit spreads, which indicate higher credit risk, are associated with deterioration from calm to moderate stress, whereas low credit spreads coincide with normalization from moderate stress to calm.

Table 24: Deviations between Covariate-Conditioned and Baseline Probabilities

Covariate	Bin	State i	N	Mean $\|\Delta\|$	Max $\|\Delta\|$	Δp_{i0}	Δp_{i1}
$c_t^{baa,10y}$	High	0	52	0.0378	0.0568	-0.0527	0.0568
$c_t^{baa,10y}$	Low	0	196	0.0100	0.0151	0.0140	-0.0151
$c_t^{baa,10y}$	High	1	175	0.0133	0.0200	-0.0200	0.0187
$c_t^{baa,10y}$	Low	1	48	0.0485	0.0728	0.0728	-0.0683
$c_t^{baa,10y}$	High	2	9	0.0000	0.0000	0.0000	0.0000
$s_t^{10y,3m}$	High	0	37	0.0046	0.0069	-0.0028	0.0069
$s_t^{10y,3m}$	Low	0	211	0.0008	0.0012	0.0005	-0.0012
$s_t^{10y,3m}$	High	1	194	0.0004	0.0007	-0.0005	-0.0002
$s_t^{10y,3m}$	Low	1	29	0.0030	0.0045	0.0031	0.0014
$s_t^{10y,3m}$	High	2	7	0.0529	0.0794	0.0000	-0.0794
$s_t^{10y,3m}$	Low	2	2	0.1852	0.2778	0.0000	0.2778

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